Supplemental Material for "Localized in-gap eigenstates of a 10^4-vertex disordered photonic network from direct interior Maxwell eigensolves"

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CONTENTS

SCOPE, REPRODUCIBILITY AND CONVENTIONS

This Supplemental Material is the complete technical record of the two computations behind the main text: the bottom-up $N = 1000$ project (delivered 2026-08-14) and the interior $N = 10^4$ project (2026-08-17 to 2026-08-31). Every number quoted in either document is listed in `report/numbers.json` of the repository with the data file it was derived from and the script that derived it; where a number can be recomputed from saved fields or spectra it *was* recomputed for this paper (scripts `report/scripts/s01`–`s06`), and the recomputed value is the one printed. Timings, operator-application counts and bake-off metrics are cited from the run records (`results/*/interior_report.json`, `results/exp/*.json`, logs). The pre-registered plans, the four adversarial-verification records and the gate ledger (`results/gates/gate_results.json`) are part of the repository. Nothing was re-solved for this paper; all analysis is CPU post-processing of saved data.

Units. $\lambda = (\omega/c)^2$ in μm^{-2}; $\nu = \omega a/2\pi c = \sqrt{\lambda}\,a/2\pi$ with $a = L/5 = 2.288\,\mu\text{m}$ for $N = 1000$ and $a = L/1250^{1/3} = 2.288\,\mu\text{m}$ for $N = 10^4$ (the same density-equivalent 8-vertex cell). Relative gap widths are $\Delta\nu/\nu = 2(\omega_2 - \omega_1)/(\omega_2 + \omega_1)$. Band numbering is MPB's: bands 1–2 are the $\omega = 0$ modes at Γ, so the n-th eigenvalue returned by the transverse solver is band $n + 2$. A ± 2 ambiguity of this convention against the original cluster montage's labelling is undecidable from local artefacts and is carried as an open item. All residuals are relative, $\|\Theta x - \lambda x\|/\|\Theta x\|$.

STRUCTURE GENERATION AND RASTERIZATION

Networks

The $N = 1000$ network is the reference LSU structure of Ref. [1] (its gap between bands $N/2$ and $N/2 + 1$ follows the band arithmetic of the parent srs/gyroid crystal [2]) as distributed with the parent project (file `N1000_lsu_example_ends.txt`: 1653 rod rows, i.e. the 1500 edges of a trivalent 1000-vertex network plus periodic duplicates of face-crossing segments; mean rod length $0.800\,\mu$m, median $0.801\,\mu$m; $L = 11.44\,\mu$m). The $N = 10^4$ network (`Structures/20260701_N10000_lsu_generated.txt`: 15 704 rows, mean rod length $0.801\,\mu$m) was generated with the same protocol at the same vertex density, $L = (N/1000)^{1/3} \times 11.44\,\mu\text{m} = 24.6467\,\mu$m ($L_{10^4}/L_{10^3} = 2.15443 = 10^{1/3}$). The LSU family constant $d_0 = 0.8\,\mu$m is the target rod length.

Rasterizer (montage convention)

`eigenfns/structure.py::rasterize_penlike` reproduces the parent notebook's `create_permittivit` algorithmically: right circular cylinders of radius r are built in an unwarped space ($x, y, z' = z/s$) and the global warp $z = s\,z'$ stretches them along z, so the cross-section perpendicular to a rod at angle θ from $\hat{z}$ is an ellipse with semi-axes r and $r\sqrt{\cos^2\theta + s^2 \sin^2\theta}$ (the direct-laser-writing "pen" sweep; near-vertical rods stay circular). Membership is tested at voxel centres with flat caps ($0 \le t \le \ell$ along the axis); a voxel is ε_{rod} if its centre lies inside any rod, else 1—binary voxels, no subpixel averaging (MPB-style smoothing [3, 4] is deliberately not applied, for fidelity to the reference montage). The implementation is bit-identical to the notebook's at 64^3 (0 differing voxels) and differs in 5 of 16.7×10^6 voxels at 256^3 (float32 vs float64 boundary arithmetic).

Decorations: *elliptical* (montage convention) $r = 0.2252\,\mu$m, $s = 2.5$, $n = 2.9275$, $\varepsilon = 8.5703$: ff $= 0.2173$ (64^3), 0.2172 (128^3), 0.2172 (256^3). *Circular* $r = 0.331836\,\mu$m, $s = 1$, $n = 2.9$, $\varepsilon = 8.41$: the radius was bisected on the $N = 10^4$ structure to ff $= 0.2200$ at 256^3 (terminating at $|\text{ff} - 0.22| = 4 \times 10^{-6}$); it gives 0.22011 at 192^3, 0.22006 at 160^3, and 0.2191 on the $N = 1000$ structure at 128^3 (gate I7: $22.0 \pm 0.5\%$, PASS). Voxel sizes: $0.0894\,\mu$m ($N = 1000$, 128^3; 11.2 vox/μm; circular rod diameter 7.4 voxels, elliptical minor diameter 5.0) and $0.1284\,\mu$m ($N = 10^4$, 192^3; 7.79 vox/μm, 70% of the $N = 1000$ resolution; rod diameter 5.2 voxels).

The boundary seam and its fix

The rod files list periodic duplicates only for segments that *cross* a face; a rod whose *radius* pokes through a face is not wrapped, so its wrap-image voxels are missing. Measured on the $N = 10^4$ 192^3 grid: filling fraction by depth from a face $d = 0, 1, 2$: 0.1975, 0.2144, 0.2201; interior ($d \geq 10$): 0.2211—an 10.7% material deficit in the outermost shell (11.8% at 256^3). The $N = 1000$ grids carry the same seam (12.8% elliptical, 11.3% circular at 128^3). In a gapped medium this thin low-index sheet is a planar defect and hosts defect states: four of the ten $N = 10^4$ in-gap states carry 18.2, 23.6, 43.7 and 42.1% of their $\varepsilon|E|^2$ in the outer 2-voxel shell (6.12% of the volume; enhancements 3.0–7.1×), and three of them peak at a voxel with two coordinates on box faces. Twenty of the 133 window modes have shell enhancement > 2; the bulk-localized in-gap candidates have 2.5–10.5% (0.4–1.7×). The $N = 1000$ windows also contain modes with shell enhancement up to 3.95× (elliptical) and 3.43× (circular), although both $N = 1000$ gaps are empty (Fig. S1).

The fix, `periodic=True`, assigns voxels by minimum image so each rod contributes its full volume wherever it sits. It changes 5528 voxels of the 192^3 grid (0.078%), every one within 3 voxels of a face; the outer-shell ff becomes 0.2177 and the global ff 0.22089 (0.22058 for $N = 1000$ circular at 128^3, 0.15% of voxels). It is a convention change relative to the reference montage and lives behind a flag; the production window was solved with the montage convention, and the gap window was *re*-solved with the fix (Sec.).

OPERATOR

$\Theta\boldsymbol{H} = \nabla \times \varepsilon^{-1}\nabla \times \boldsymbol{H}$ in MPB's transverse plane-wave representation [5]: every field is a pair of complex amplitudes on the two transverse unit vectors $(\hat{t}_1, \hat{t}_2) \perp (\boldsymbol{k} + \boldsymbol{G})$ for every reciprocal vector on the grid, so $\nabla \cdot \boldsymbol{H} = 0$ is exact by construction and the curl is the 2×2 block $|\boldsymbol{k} + \boldsymbol{G}|\, \sigma_y$ on each plane wave. One application: spectral curl → Cartesian → inverse FFT → multiply by $\varepsilon^{-1}(\boldsymbol{r})$ → FFT → project the curl back: six 3-D FFTs. At $k = \Gamma$ the $\boldsymbol{G} = 0$ plane wave has no transverse pair; both its amplitudes are zeroed, which removes the two $\omega = 0$ modes exactly and makes Θ positive definite on the remaining space (dimension $2G^3 - 2$). With real scalar ε the operator is real symmetric on real fields. Validation of the operator itself: against the analytic homogeneous spectrum at $G = 8$ the

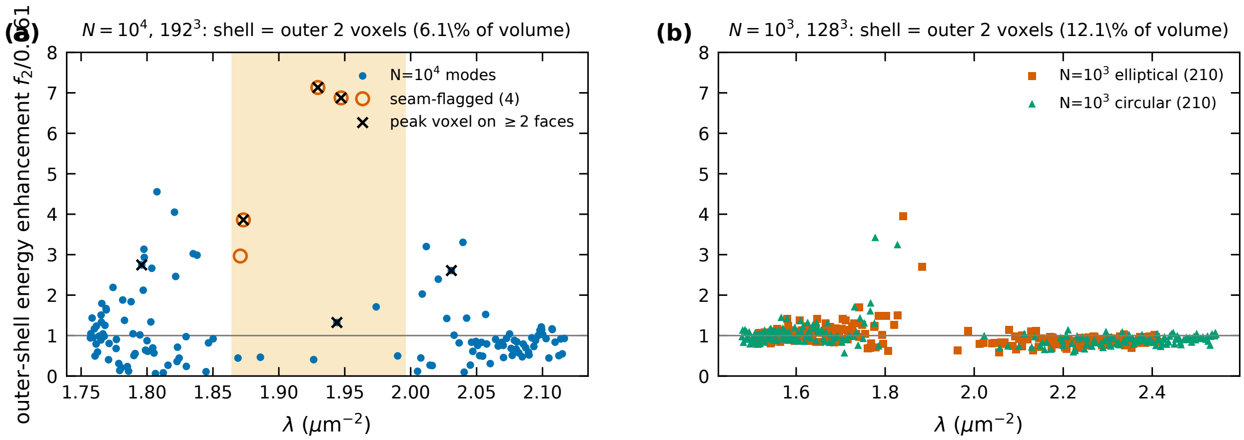

FIG. S1. Outer-2-voxel-shell energy fraction divided by the shell volume fraction for every window mode. (a) $N = 10^4$, 192^3: the four seam-flagged in-gap states (open red) and the modes whose peak voxel has two coordinates on a face (crosses). (b) The two $N = 1000$ windows at 128^3 (shell volume 12.1%).

TABLE S1. Operator application time and spectral extent per grid ($N = 10^4$, circular decoration; `results/exp/n10k_G*_timing.json`).

grid	vox/μm	Θ apply (ms/vector)	$\lambda_{\max}$ (μm^{-2})	vector (GB)
192^3	7.79	23.2	2209	0.113
224^3	9.09	38.2	3031	0.180
256^3	10.39	53.9	3975	0.268
288^3	11.69	79.2	5056	0.382

maximum relative error is 2.6×10^{-7} (complex64) with null dimension exactly 2; against an independent dense fp64 three-component construction the nonzero spectra agree to 1.2×10^{-7} and the Hermiticity residual is 5.6×10^{-8}.

Measured cost (complex64, RTX 4080 Laptop 12 GB, chunked application): 2.7 ms/vector at 128^3 ($N = 1000$); 23.2 (192^3), 38.2 (224^3), 53.9 (256^3), 79.2 ms (288^3) for $N = 10^4$ (Table S1). $\lambda_{\max}$ (two-seed Lanczos upper estimate $\times 1.05$): 4633.6 (128^3, $N = 1000$), 2208.9 (192^3), 3030.7 (224^3), 3974.6 (256^3), 5055.9 (288^3). One vector is $2G^3$ complex64 $= 0.113$ GB at 192^3, 0.268 GB at 256^3.

BOTTOM-UP SOLVER ($N = 1000$)

Algorithm

Deflated block LOBPCG [6, 7] (`eigenfns/solver.py`): blocks of $m = 32$ with 12 guard vectors, warm-started from the previous block's guard Ritz vectors; MPB's transverse-projection preconditioner [5] (Eq. 14 there; the same six-FFT cost as Θ; it reduced per-block iteration counts to 18–72 and made them resolution-stable); SVQB orthonormalization with drop threshold 10^{-5} (the fp32 Gram noise floor); all small dense algebra (Gram eigendecompositions, Rayleigh–Ritz) in fp64 on the host; ΘX recomputed every iteration (tracked products drift by $\sim 10^{-3}\|\Theta W\|$ in complex64); re-deflation against the locked set every iteration; rank-dropped rows penalized on the Rayleigh–Ritz diagonal; fixed array shapes throughout (XLA recompiles otherwise); chunk-assembled Rayleigh–Ritz; host-resident locked set streamed in ≈ 0.75 GB chunks with a barrier per chunk; per-block checkpointing with automatic resume. Locking tolerance: relative residual $\leq 10^{-4}$ (Weyl: $\Delta\omega/\omega \leq 5 \times 10^{-5}$ unconditionally; the measured interior error is $\sim 4 \times 10^{-6}$).

TF32. On Ada GPUs XLA's default runs fp32 matmuls in TF32 (10-bit mantissa); measured at 64^3 the Gram/rotation accuracy floors at $\sim 10^{-3}$ and block residuals plateau at $\sim 7 \times 10^{-3}$. Every matmul/tensordot carries `precision=HIGHEST`. FFTs are unaffected. A second precision trap, found in adversarial round 3 of the interior project, is described in Sec. .

Gates G1–G9

Pre-registered on 2026-08-13 (thresholds verbatim in `plans/2026-08-13_preregistered_plan.md`); outcomes in Table S2. The judge protocol for parity is: MPB 1.11.1 reads our ε grid from file (`epsilon-input-file`), we read back MPB's `-epsilon.h5` and solve on *that* grid, so the two codes see the identical discrete problem.

The gap at 64^3 depends on the ε representation

A new observation made while recomputing the parity numbers: on the MPB judge grid (MPB's trilinear read-back of our binary 64^3 grid), both codes agree to 3.5×10^{-5} but

TABLE S2. $N = 1000$ gates (pre-registered criteria; recomputed values where the data allow).

gate	registered test / threshold	outcome	measured
G1 crystal parity	srs cell + $2 \times 2 \times 2$ supercell, Γ + 3 k-points vs MPB, $\Delta\omega/\omega \leq 10^{-4}$	**not run**	never executed as registered; production claims are Γ-only; no $k \neq \Gamma$ validation exists. The identical-grid protocol was validated on the disordered structure instead (G3)
G2 literature	srs at Sellers's parameters, gap 28.06 ± 1.5 pp	PASS	27.97%
G3 parity	300 bands, 64^3, $\Delta\omega/\omega \leq 10^{-4}$ per band	PASS	298 bands: max 8.95×10^{-6}, median 1.42×10^{-6} (MPB tol 10^{-7}, fp64; 2 h 22 m CPU vs 33 min for our solver on CPU)
G3w parity	660 bands, 64^3	PASS	658 bands: max 3.55×10^{-5}, median 1.39×10^{-6}; +2 band alignment unique
G4 degeneracies	clusters ($\Delta\lambda < 10^{-3}\lambda$) compared as subspaces, principal-angle cosine ≥ 0.99	PASS (scope-limited)	6 low-lying clusters (bands $\leq$ 30), min cosine 0.999986; window clusters (64% of adjacent pairs qualify) were not field-compared
G5 convergence	$\omega(G)$ for bands {398,450,500,501,550,607} at 96/128/160^3 (160^3 not run; 64^3 substituted) and ω(tol) sweep (not run); monotone	**FAIL** as registered	5/6 bands monotone, worst Richardson residual 0.23% at 128^3; band 500 non-monotone, spread 0.274% (55× the tol-10^{-4} Weyl bound, 1800× the measured solver–MPB error): rasterization-limited (Fig. S2)
G6 completeness	(a) monotone locking; (b) deflated-probe KPM count of missed eigenvalues, $= 0$ with s.e. ≤ 0.5	PASS below the gap (amended)	(a) zero violations; (b) at mid-gap, degree 8000: 0.21 ± 0.02 (Jackson-tail bias, inconsistent with a missed band). First version (threshold at band 619, degree 800) read 86.6 phantom missing bands—a smearing artefact, recorded. Above-gap completeness rests on monotone locking + 64^3 parity
G7 residuals	rel-res $\leq 10^{-4}$; orthonormality $\leq 10^{-3}$	PASS	worst 9.78×10^{-5}; full 210×210 window Gram $\lvert G - I\rvert_{\max} = 1.2 \times 10^{-5}$; window-vs-

TABLE S3. Parity and precision tests (recomputed from the saved spectra; $\Delta\omega/\omega$ unless stated).

test	grid / bands	n	max $\Delta\omega/\omega$	median
MPB parity (32^3, 150 bands, tol 10^{-9})		148	4.3×10^{-6}	8.6×10^{-7}
G3: MPB parity (64^3, 300 bands, tol 10^{-7})		298	9.0×10^{-6}	1.4×10^{-6}
G3w: MPB parity (64^3, 660 bands)		658	3.5×10^{-5}	1.4×10^{-6}
G9: c64 vs c128 (48^3, 96 bands)		96	2.4×10^{-7}	–
I1: interior vs bottom-up ($N = 1000$, 128^3; $\Delta\lambda/\lambda$)		55	2.4×10^{-7}	6.3×10^{-8}
I4: interior vs bottom-up ($N = 1000$ circ.; $\Delta\lambda/\lambda$)		216	4.1×10^{-7}	9.8×10^{-8}

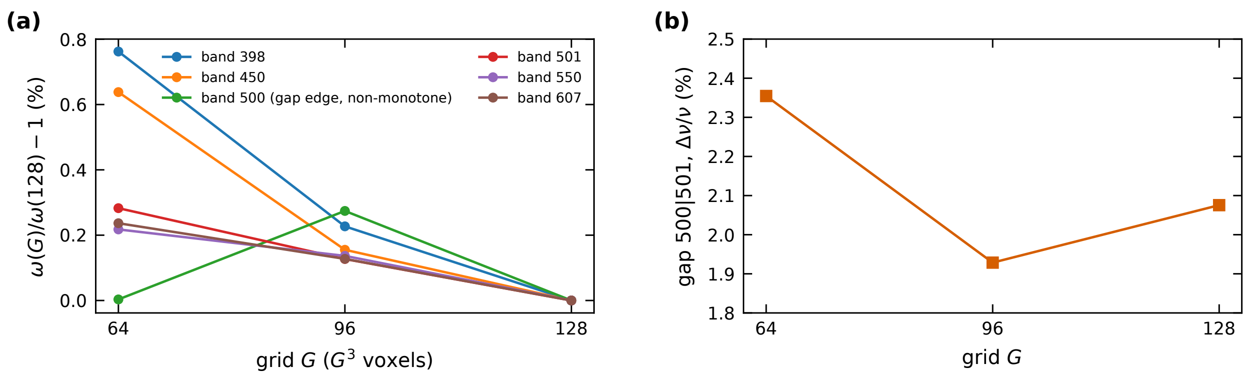

FIG. S2. G5: (a) $\omega(G)/\omega(128^3) - 1$ for the six registered bands at $64^3/96^3/128^3$; band 500 (gap edge) is non-monotone. (b) The elliptical gap width vs grid.

the elliptical 500|501 spacing is only 0.60% and the largest spacing in the window sits at 501|502 (0.83%), whereas on our binary 64^3 raster the gap is at 500|501 with 2.35% (501|502: 0.61%). Bands 498–503 differ between the two ε representations by −1.4% to −3.3%. The binary raster is the montage convention and is what all production results use; at 128^3 the 500|501 spacing is 9.6× the median of its twenty neighbours (elliptical) and 26× (circular), so the statement is robust there, but at 64^3 "the gap is at 500|501" is a statement about the binary-voxel convention. This is the same rasterization sensitivity that failed G5.

Performance

611 bands at 128^3 (elliptical): 19 908 s = 5 h 32 m, 31 blocks with 23–72 iterations each, $\approx 1.3 \times 10^5$ operator applications; the registered target was 4 h (missed; inside the 12 h

cap). Circular decoration, same window: 8876 s, 6.97×10^4 applications. 96^3: 6426 s. 680 bands at 64^3: 942 s. Against MPB at matched work (300 bands, 64^3): $\sim 18\times$ in wall clock, with stated confounds (MPB's tolerance 10^{-7} vs our 10^{-4}, fp64 vs complex64, single process vs a 32-thread host); the tolerance-matched, core-for-core advantage is likely single-digit.

INTERIOR SOLVER ($N = 10^4$)

Why new machinery

Extrapolating the production profile to the ≈ 5150 bands below the $N = 10^4$ gap at 256^3: locked-set storage 5150×0.268 GB $= 1.38$ TB ($12\times$ RAM + disk), deflation time $O(100)$ days (order-of-magnitude; the n^2 regime is not yet visible at $N = 1000$ scale), pure matvec time 16.4 h. A bottom-up Chebyshev filter hits the same storage wall.

Kernel polynomial method

Stochastic Chebyshev moments $\mu_k = \langle z|T_k(B)|z\rangle$ with $B = (2\Theta - \lambda_{\max})/\lambda_{\max}$, Rademacher probes z restricted to the transverse space, even/odd doubling (degree-p moments from $p/2$ applications), Jackson kernel [8]. Counting function $N(\lambda_b) = \sum_k g_k \mu_k c_k(x_b)$ with c_k the Chebyshev coefficients of the step; per-probe values give the standard error. Jackson width at position λ: $\sigma(\lambda) = \pi\sqrt{1-x^2}\,\lambda_{\max}/2p$.

Validation on $N = 1000$ (same 128^3 grid as the exact 611-band spectrum; degree 8000, 16 probes, 731 s): window count 213.3 ± 2.6 vs exact 210; count below mid-gap 501.5 ± 2.7 vs exact 498; total-trace normalization exact. The gap edges from the 10%/20% criteria, $[1.861, 1.974]/[1.830, 2.013]$ on a uniform 5×10^{-4} grid, bracket the exact $[1.883, 1.963]$ within the smearing width 0.037, i.e. KPM edges carry a criterion-dominated systematic of ± 0.02–0.03 (Fig. S3).

$N = 10^4$ (256^3, degree 12 000, 12 probes, $\lambda_{\max} = 3974.6$, 8477 s; Jackson width at the gap 0.023): the count below $\lambda = 1.957$ is 5010.3 ± 12.5 (0.501 bands/vertex; 5008.3 ± 12.6 below the bracket centre 1.930), placing the gap between MPB bands $\approx 5012|5013 \pm 13$. The registered nominal gap is the interval where the smoothed DOS is below 160 states per unit λ, $[1.864, 1.996]$ ($\Delta\nu/\nu = 3.41\%$); the 5% and 20% criteria of the project (thresholds 80 and 320) give $[1.885, 1.968]$ (2.14%) and $[1.837, 2.022]$ (4.78%). [The project's script normalizes by a

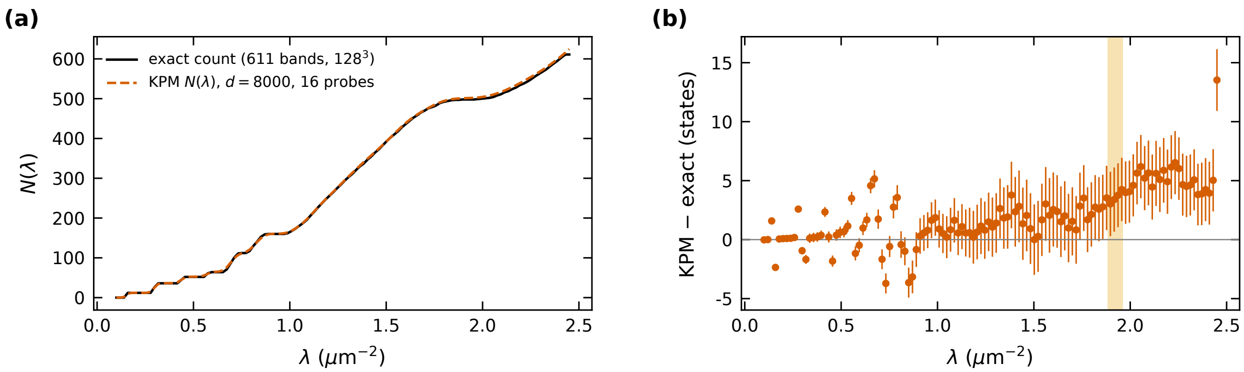

FIG. S3. KPM validation on $N = 1000$: (a) counting function vs the exact 611-band spectrum; (b) difference (states), with the exact gap shaded.

"local median" DOS whose grid dependence we could not reproduce exactly; the registered brackets are reproduced to the grid step by the absolute thresholds, which is how we state them.] Window populations: 139.3 ± 2.8 in the production window $[1.757, 2.117]$, 317.1 ± 5.1 in $[1.71, 2.19]$, 69.2 ± 1.5 / 66.1 ± 2.4 in $S_{\text{below}}/S_{\text{above}}$, 4.87 ± 0.41 in $S_{\text{gap}} = [1.925, 1.985]$, 11.18 ± 0.58 in the bracket, 3.80 ± 0.38 in the certifying sub-interval $[1.9063, 1.9606]$. These are stochastic errors only: the Jackson-smoothed count of an interval is biased by $\approx (\sigma^2/2)\rho'$ at each edge, and both production-window edges sit on steep shoulders ($\rho = 1281$ and 996 states per unit λ, $\rho' = -1.9 \times 10^4$ and $+1.1 \times 10^4$), giving a predicted bias of +7.7 states (project record: $+7.5 \pm 0.3$), so the 139 vs 133 comparison is *not* a completeness statement (Sec.); on the $N = 1000$ calibration six nested gap-straddling windows all have positive error. The same bias makes the absolute tile labels 4942–5074 read ≈ 4.5 low; the count of 133 is exact, the absolute index is not certified beyond ± 13.

Bandpass filter and two-stage subspace iteration

The bandpass filter is the Jackson-damped Chebyshev expansion of the indicator $1_{[\lambda_{\text{lo}}, \lambda_{\text{hi}}]}$, built as the difference of two step expansions and accumulated during the three-term recurrence (`eigenfns/interior.py::bandpass_coeffs`, `_apply_cheb_poly`): $Y = \sum_k c_k T_k(B) X$, one Θ application per degree, chunk re-normalized. The transition width scales as $\delta\lambda_t \approx \pi\sqrt{\lambda\lambda_{\max}}/p$; at 192^3 this is ≈ 0.017 at degree 12 000 (≈ 27 states per edge in the bulk) and ≈ 0.0086 at 24 000. Two stages: *build* at degree 3000–4000, two outer iterations from random starts with $m = 1.5\times$ the KPM-predicted population, then *polish* with the same filter

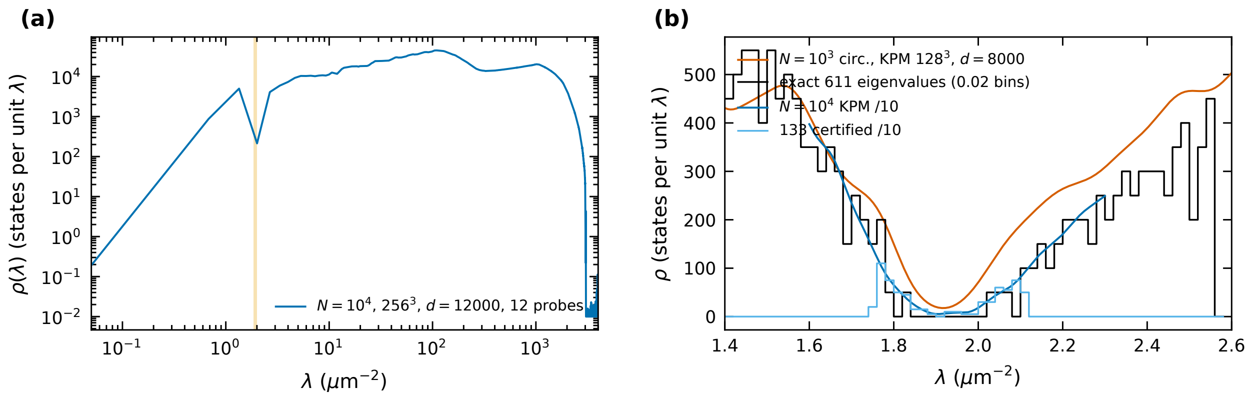

FIG. S4. (a) Full-bandwidth KPM DOS of the $N = 10^4$ structure. (b) The gap region: $N = 1000$ circular KPM and exact histogram; $N = 10^4$ KPM and the histogram of the 133 certified eigenvalues (both scaled by 1/10).

at degree 12 000 on the (trimmed) basis, up to 4–6 outer iterations; the measured residual damping per polish outer was $\times 12$ at production against $\times 0.62$ for an undersized basis at low degree, which is why two stages beat one. The basis is host-resident and streamed in $\leq$ 8-vector chunks through every stage (filter, SVQB, Rayleigh–Ritz assembly, rotation, residuals); the hosted path matches the device path to $\Delta\lambda \leq 4.5 \times 10^{-8}$ and costs $\approx 1.5\times$ the raw matvec time. Extraction is exclusively `rr_extract`: SVQB orthonormalization, fp64 Rayleigh–Ritz on the *original* Θ, per-pair residuals on Θ; pairs failing 10^{-4} are never reported. Ghosts are therefore excluded by construction; zero ghosts were observed in every run of every method. Cross-slice duplicates are identified by eigenvector overlap (> 0.5), never by eigenvalue proximity.

Bake-off

On a 50-band interior slice of the exact $N = 1000$ spectrum (0-based indices 473–522 = MPB bands 476–525, $\lambda \in [1.711, 2.146]$, $\sigma = 1.9229$; relative slice width 9.4×10^{-5} of $\lambda_{\max}$, close to the $N = 10^4$ ratio 1.2×10^{-4}) four arms were run at 128^3 (Table S4, Fig. S5). Folded-spectrum LOBPCG on $(\Theta - \sigma)^2$ with the Wang–Zunger squared-kinetic preconditioner [9, 10] locked 20 vectors at maxit 200 with folded Rayleigh quotients $\mu \in [0.096, 0.279]$ while all 50 targets have $\mu \leq 0.0497$: no preconditioner grips the ε-structure-dominated low spectrum (the ESCAN kinetic-diagonal analogy does not transfer); an earlier residual-based diagnosis

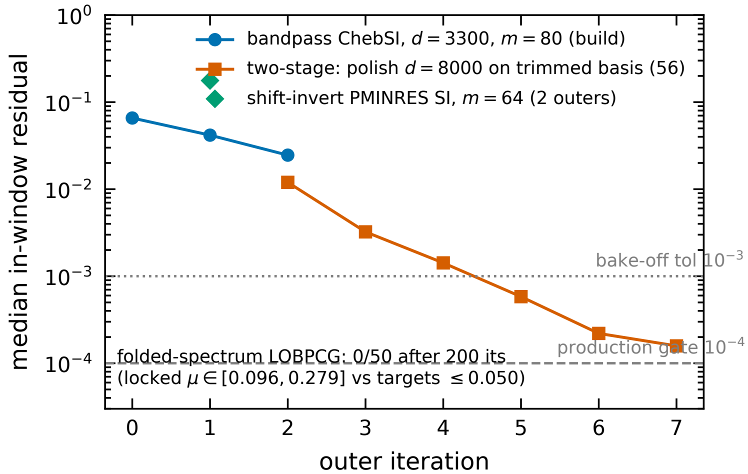

FIG. S5. Bake-off: median in-window residual vs outer iteration.

of this failure was itself refuted in adversarial round 1 and replaced by the μ-trajectory argument. Shift-invert subspace iteration with preconditioned MINRES inner solves was tolerance-starved (inner iterations at maxit 150/150; median residual 0.18), reproducing the published verdict of Ref. [10]. One-stage bandpass at degree 3300 captured the subspace (50/50 within 2×10^{-3}) but damped residuals only $\times 0.62$ per outer. (A filtered-Lanczos alternative [11] was not run: its stored basis of 2.5–5$\times$ the target count exceeds host memory at 192^3.) Two-stage bandpass verified 24 pairs at 10^{-3} within the budget, the remaining 26 (slice-edge and gap-edge pairs) still descending; the lesson (basis must be $\geq 1.5\times$ the population) set the production oversampling. Five variants of an expansion-Rayleigh–Ritz polish (LOBPCG-style, filtered residual expansion) were unstable at scale (interior-Ritz ghost instability) and abandoned.

The Rayleigh-normalization correction

Adversarial round 3 (2026-08-25) found that `rr_extract` reported the Ritz value as $\langle x, \Theta x\rangle$ without dividing by $\|x\|^2$, while SVQB in blocked fp32 leaves $\|x\|^2 - 1 = +2.8 \times 10^{-5}$ (128^3) and $+4.7 \times 10^{-5}$ (192^3; range 4.4–6.5 $\times 10^{-5}$ over the 133 modes)—blocked fp32 accumulation over millions of positive terms under-estimates the Gram diagonal, so the vectors come back slightly over-normalized. Every eigenvalue was biased high by that relative amount. The fp64 check shows the eigenvalue error correlated with $\|x\|^2 - 1$ at $r = 0.969$ (Fig. S6b), and dividing it out improves the I1 ground-truth parity from 2.85×10^{-5} to

TABLE S4. Bake-off on the 50-band $N = 1000$ slice (`results/exp/bakeoff_*.json`, `.log`). "Verified" = residual $\leq 10^{-3}$ on Θ and matched to a reference eigenpair. The two-stage wall time is the polish legs; its build is the one-stage row.

Method	Configuration	verified	ghosts	Θ-applies	wall (s)	Failure mode / note
Bandpass ChebSI, one stage	$m = 80$, $d = 3300$, 3 outers	0/50	0	792,528	7,109	subspace captured 50/50 to 2×10^{-3}; med. res. $6.6 \to 4.2 \to 2.5 \times 10^{-2}$
Two-stage bandpass (build + polish)	build $d = 3300$ $m = 80$ $\times 2$, trim 56, polish $d = 8000$ $\times 6$	24/50	0	3,217,088	18,464	26 unconverged = slice-edge/gap-edge pairs still descending; trim-limited (1.12× oversampling)
Shift-invert PMINRES SI	$m = 64$, inner tol 10^{-2}, maxit 150, 2 outers	0/50	0	32,336	441	inner solves hit maxit (150/150); med. res. 1.8×10^{-1}
Folded-spectrum LOBPCG	$m = 32$, WZ preconditioner $\alpha^2 \in \{0.05, 1, 20\}\sigma^2$	0/50	0	–	629	first block locked 20 at maxit 200 with folded $\mu \in [0.096, 0.279]$ vs targets $\mu \leq 0.0497$
Expansion-RR polish (5 variants)	on the two-stage build subspace	0/50	0	–	–	med. res. $4.2 \times 10^{-2} \to 0.9$ in one sweep; continuity/strip variants oscillate 7–23 $\times 10^{-3}$

2.35×10^{-7} (121×). The trap: the same check in fp32 gives the wrong sign and magnitude (-3.4×10^{-4}), because an fp32 sum over 4.2×10^6 terms carries its own $\sim 10^{-4}$ error. The code was fixed, and all saved eigenvalues were corrected retroactively as $\lambda_{\mathrm{corr}} = \lambda_{\mathrm{raw}}/\|x\|^2$ with fp64 norms of the saved vectors (`scripts/exp/fix_rayleigh_norm.py`; originals kept as `window_eigenvalues_raw.npy`, norms as `window_norms.npy`); this is exactly the Rayleigh quotient of the same vector, so no re-solve was needed. Effects: a uniform shift of -4.7×10^{-5} relative (9×10^{-5} absolute at $\lambda \approx 1.94$) against a median level spacing of 1.4×10^{-3}; no gap, spacing, localization or DOS comparison moves. The reported residuals were computed

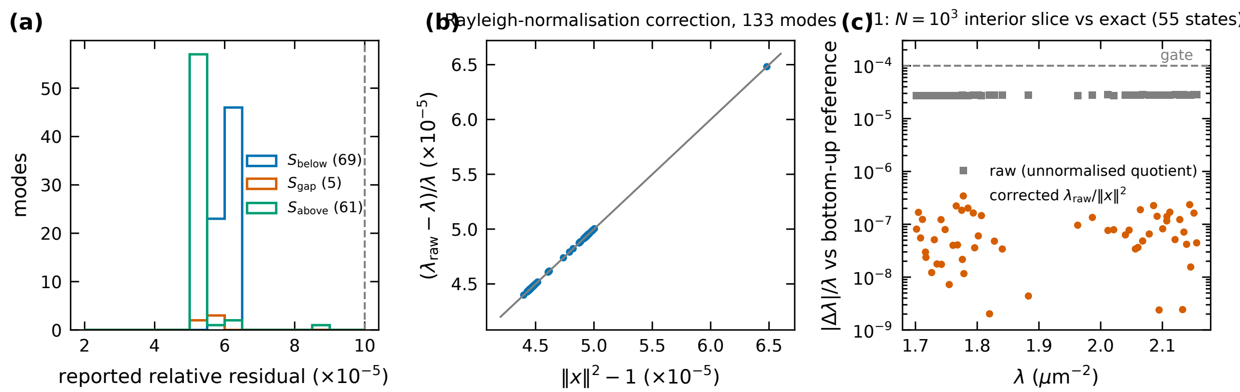

FIG. S6. (a) Reported residuals of the three production slices. (b) Relative eigenvalue shift of the correction vs $\|x\|^2 - 1$ (line: identity). (c) I1: per-state $|\Delta\lambda|/\lambda$ of the interior slice against the exact bottom-up eigenvalues, raw and corrected.

with the raw λ and carry a floor of the same size ($r^2_{\mathrm{rep}} = r^2_{\mathrm{true}} + \delta^2$): the worst reported residual 8.7×10^{-5} is $\approx 5.9 \times 10^{-5}$ after the correction (direct fp64 recomputation in the ledger: 5.88×10^{-5}), so about half the residual budget was being spent on the bug; no state was lost to it (zero in-window unconverged pairs in all three slices). The correction was independently confirmed twice: a fresh interior solve of the circular $N = 1000$ decoration with the fixed code lands at 4.1×10^{-7} against a different bottom-up reference; and the two states found by two different slices, whose raw eigenvalues differed by 1.42×10^{-6} and 1.73×10^{-6}, agree to 7.7×10^{-9} and 1.2×10^{-8} after dividing each by its own $\|x\|^2$.

Production runs

Table S5 lists every interior run. The production window $[1.757, 2.117]$ was descoped at registration from the ~ 300-band window $[1.71, 2.19]$ (10–15 GPU-days at 256^3) to ≈ 139 bands at 192^3; it was solved as $S_{\mathrm{below}} = [1.757, 1.930]$ ($m = 104$) and $S_{\mathrm{above}} = [1.980, 2.117]$ ($m = 100$), plus the Amendment-A2 slice $S_{\mathrm{gap}} = [1.925, 1.985]$ ($m = 16$) added after the first in-gap states appeared. $69 + 5 + 61 = 135$ pairs, 133 after removing the two states found twice (eigenvector overlaps 0.9988 and 0.9988; the second slice's copy is dropped). Every slice reports zero in-window unconverged pairs. Total: 1.16×10^7 operator applications, 104.3 GPU-hours. Every run detaches with per-outer rolling checkpoints; 5+ external kills, one host crash and one self-inflicted race each cost at most one outer iteration.

TABLE S5. All interior runs (`results/*/interior_report.json`). $\lambda_{\max}$ and Θ counts of the 256^3 low-edge anchor were lost in a post-solve OOM and are "n/a"; its eigenpairs were never at risk.

run	grid	window (μm^{-2})	m	d_{build}	d_{polish}	conv.	unconv.	Θ-applies	wall (h)	worst res.
S_{below}	192^3	[1.757, 1.930]	104	3000	12000	69	0	6.87M	63.3	6.4×10^{-5}
S_{gap}	192^3	[1.925, 1.985]	16	4000	12000	5	0	0.51M	3.6	5.8×10^{-5}
S_{above}	192^3	[1.980, 2.117]	100	3000	12000	61	0	4.20M	37.5	8.7×10^{-5}
periodic re-solve v1	192^3	[1.855, 2.000]	30	4000	12000	7	1	2.04M	20.0	5.9×10^{-5}
periodic re-solve v2	192^3	[1.855, 2.000]	48	4000	12000	7	2	2.69M	4.0	3.5×10^{-5}
I6 160^3 leg	160^3	[1.800, 2.060]	72	3000	10000	45	18	6.19M	18.3	8.2×10^{-5}
I6 256^3 low-edge anchor	256^3	[1.840, 1.950]	18	4000	16000	11	0	n/a	27.2	9.8×10^{-5}
I6 256^3 high-edge anchor	256^3	[1.990, 2.035]	18	4000	16000	3	8	1.30M	26.8	1.0×10^{-4}
I1 slice ($N = 1000$, 128^3)	128^3	[1.701, 2.156]	80	3000	8000	55	1	4.32M	2.6	5.2×10^{-5}
I4 below ($N = 1000$ circ.)	128^3	[1.470, 1.835]	155	2500	8000	107	0	2.02M	4.0	9.6×10^{-5}
I4 above ($N = 1000$ circ.)	128^3	[2.015, 2.550]	160	2500	8000	109	0	3.36M	6.5	3.5×10^{-5}

Completeness

The registered estimator (I2 v1: two independent `kpm_count_below` calls) was misdesigned—each call counts ~ 5000 states with stochastic error ~ 26 states, and it placed the locked set on the GPU (OOM); its one execution (missed $= 0.66 \pm 26.5$) is recorded as a failure. Amendment A3 replaced it by a single Chebyshev recurrence per probe chunk accumulating the Jackson bandpass $\rho(\Theta)$ of the window with probes deflated against the converged vectors: $E[z^\dagger \rho(\Theta) z] = \sum_{\text{missed}} \rho(\lambda_i) + \mathcal{L}$, where $\mathcal{L}$ is the transition-zone leakage of states outside the window. $\mathcal{L}$ is (A) predicted from the KPM DOS, or (B) made negligible by evaluating on a sub-interval whose edges sit in the sparse gap region. A3 made (B) the certifying test and (A) a consistency check *before* either ran. Results (degree 24 000, 12 probes): sub-interval $[1.9063, 1.9606]$, missed $= -0.0002 \pm 0.0001$ (gate $|\text{missed}| < 0.5$; true leakage there 0.0017)—**PASS**; full window, missed $= +1.68 \pm 0.32$, whose leakage model the $N = 1000$ calibration (same structure, decoration, grid and $\lambda_{\max}$ for moments and solve) shows to *over*-predict by at least 1.23 ± 0.23 states, so the full-window figure must not

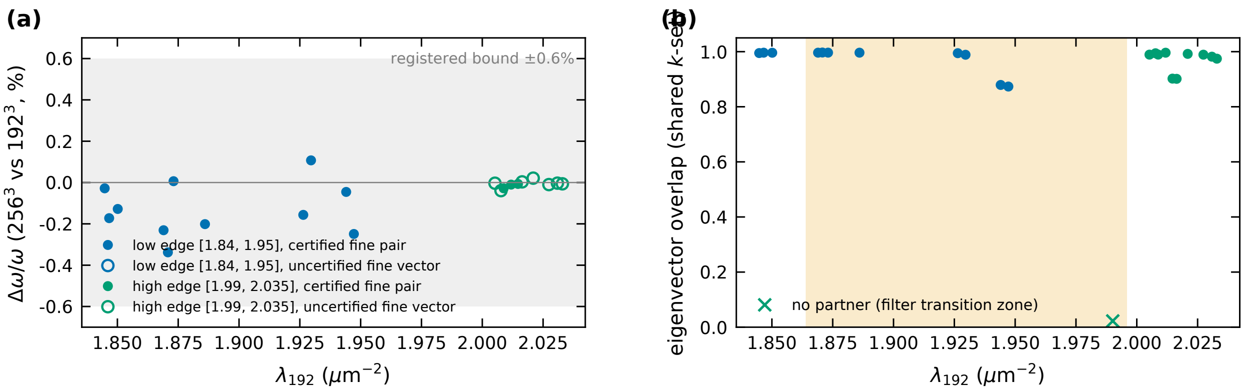

FIG. S7. I6: (a) $\Delta\omega/\omega$ between 256^3 and 192^3 per matched state; (b) overlaps.

be read as a count either way. Three defects of the leakage computation (disjoint-mask trapezoid, error propagation, double-counting of deflated states) were corrected post hoc and both versions are in the ledger. The KPM window population 139.3 ± 2.8 vs 133 found is not evidence of missing states (bias +7.7) and not evidence of completeness.

Gates I1–I9

Registered 2026-08-18 with Amendments A1 (polish cap $4 \to 6$ after the first I1 run found 49/50: a near-degenerate pair with $\Delta\lambda = 7 \times 10^{-4}$ needed outers 5–6), A2 (in-gap states: the "gap empty" clause of I5 will be reported failed; S_{gap} added), A3 (completeness estimator, above). Outcomes in Table S6: 8 PASS, 3 FAIL, the rest recorded.

The seam test

The gap window $[1.855, 2.000]$ was re-solved at 192^3 on the periodically wrapped structure twice: $m = 30$ (20.0 h; 7 converged pairs, 1 in-window unconverged at 1.9095) and $m = 48$ (4 of 8 requested polish outers, stopped on a two-outer plateau; the same 7 pairs, agreeing with the first to 4.8×10^{-5}, exactly the normalization correction that the second run had built in; 2 unconverged at 1.8902 and 1.9534 with residuals 6×10^{-2} throughout). Identity between the montage-convention and periodic states is decided by eigenvector overlap > 0.5 in the shared plane-wave basis (Fig. S8), not by a λ window (a 2×10^{-3} window mis-called four persisting states because the physical shift is larger). Results: all six bulk in-gap states persist with overlaps 0.952–0.9997 and $\Delta\lambda = -0.0007$ to -0.0034, every shift

TABLE S6. $N = 10^4$ interior gates (criteria verbatim from `plans/2026-08-18_interior_preregistration.md`).

gate	registered test / threshold	outcome	measured
I1 ground truth	production-config interior solve of the $N = 1000$ 50-band slice vs the bottom-up spectrum; all 50 found, $\Delta\omega/\omega \leq 10^{-4}$, cluster proj$^2 \geq 0.99$, 0 ghosts	PASS	50/50, 0 ghosts, worst residual 5.2×10^{-5}; max $\Delta\lambda/\lambda = 2.35 \times 10^{-7}$, median 6.3×10^{-8} over 55 pairs after the normalization correction (2.83×10^{-5} before); proj$^2 = 0.999993$ (the scorer's 1.0000203 was the same normalization artefact). First run 49/50 → A1
I2 completeness	deflated-probe KPM count ≤ 0.5	v1 FAIL (design) / v2 PASS	sub-interval -0.0002 ± 0.0001 certifies; full window $+1.68 \pm 0.32$ is a biased consistency check (Sec.)
I3 residuals & orthonormality	rel-res $\leq 10^{-4}$; window Gram $\lVert G - I \rVert_{\max} \leq 5 \times 10^{-5}$	**FAIL**	residuals PASS (worst 5.88×10^{-5}, median 3.26×10^{-5}, fp64 recomputation); Gram 4.86×10^{-4}, 9.7× the gate, entirely off-diagonal and entirely *cross-slice* (same-slice max 8.6×10^{-6}): SVQB enforces orthonormality within a slice and nothing does across the three independent solves. Six of the ten worst pairs involve the seam state 1.9472. Two explanations (small spacing; poor convergence) were refuted by measurement; mechanism open. The merged set is used only for spectra, montages and localization, which do not need mutual orthonormality
I4 new decoration	bottom-up vs interior on $N = 1000$ circular, window 398–607, same thresholds as I1	PASS	bottom-up: gap 500|501, 5.07%; interior 107/107 below, 109/109 above, 0 in-window unconverged; 210/210 vector-matched targets, 0 missed, 0 ghosts, max $\Delta\lambda/\lambda = 4.11 \times 10^{-7}$, min proj$^2 = 0.9961$; 6 overhang modes checked on eigenvalues alone ($\leq 1.9 \times 10^{-7}$)
I5 spectrum consistency	eigen-edges inside the KPM 5%→20% bracket; gap inter-	**FAIL** & as registered	ten certified pairs inside $[1.864, 1.996]$; recorded before S_{gap} ran (A2) as the physics finding, not

TABLE S7. I6 cross-grid pairing by eigenvector overlap (192^3 coefficients embedded in the 256^3 k-set by integer wavenumber; the transverse frames are identical at equal $\boldsymbol{n}$).

edge	λ_{192}	λ_{256}	overlap	$\Delta\omega/\omega$ (%)	note
low	1.84478	1.84375	0.995	-0.028	
low	1.84662	1.84026	0.996	-0.172	
low	1.85015	1.84542	0.996	-0.128	
low	1.86898	1.86036	0.996	-0.231	
low	1.87076	1.85814	0.997	-0.338	
low	1.87308	1.87333	0.996	+0.006	
low	1.88601	1.87842	0.996	-0.201	
low	1.92641	1.92039	0.994	-0.157	
low	1.92960	1.93374	0.989	+0.107	
low	1.94405	1.94228	0.879	-0.045	
low	1.94721	1.93752	0.873	-0.249	
high	1.99012	2.03410	0.022	+1.099	no partner (filter transition zone)
high	2.00521	2.00508	0.989	-0.003	unconverged fine vector
high	2.00769	2.00612	0.994	-0.039	unconverged fine vector
high	2.00879	2.00766	0.989	-0.028	
high	2.01185	2.01145	0.996	-0.010	
high	2.01468	2.01440	0.902	-0.007	
high	2.01637	2.01650	0.901	+0.003	unconverged fine vector
high	2.02093	2.02179	0.992	+0.021	unconverged fine vector
high	2.02738	2.02698	0.989	-0.010	unconverged fine vector
high	2.03088	2.03076	0.982	-0.003	unconverged fine vector
high	2.03292	2.03267	0.975	-0.006	unconverged fine vector

negative as required when the missing dielectric is added (1.8690 → 1.8683, 1.8860 → 1.8853, 1.9264 → 1.9242, 1.9441 → 1.9407, 1.9738 → 1.9708, 1.9901 → 1.9879). Three seam-flagged states have no partner (best overlaps 0.006, 0.131, 0.090; total overlap budgets $\sum_j |\langle m|p_j\rangle|^2$ of 10^{-4}, 0.020, 0.009). The fourth, 1.9472, has best overlap 0.30 and budget 0.097, 5–1000×

TABLE S8. Localization of the seven certified states of the seam-free re-solve ($m = 48$).

λ (μm^{-2})	p (%)	ξ (μm)	r^2	f_2 (%)	montage partner
1.8682	0.071	1.83	0.995	2.6	1.8690
1.8852	0.034	1.78	0.992	2.8	1.8860
1.9241	0.046	1.87	0.985	2.0	1.9264
1.9406	0.500	2.24	0.988	4.2	1.9441 (extended in the montage run; resolved here)
1.9707	0.328	2.12	0.997	10.3	1.9738
1.9839	0.587	2.55	0.994	13.1	2.0088 (pulled in from above)
1.9878	0.321	2.63	0.986	3.7	1.9901

the others: periodic state 1.9407 has 0.9969 of its weight inside span{1.9441, 1.9472}, and the I2 audit of the periodic solve finds $+2.04 \pm 0.42$ states missing in the interval containing it. Since the same two-dimensional-subspace argument is used to accept the 192/256 pairing of this very pair (overlaps 0.879/0.873, combined 0.996/0.993), it cannot be rejected here: 1.9472 is **undetermined**. The one periodic state without an in-gap partner, 1.9839, is the montage state 2.0088 pulled into the gap by the added dielectric ($\Delta\lambda = -0.0248$, overlap 0.915); all seven periodic states have a montage partner above the rule. In-gap count after the fix: $10 \to 7$. Negative half of the verdict weakened: the periodic solve is incomplete, so "no partner found" is not "vanished". The seam-free re-solve's localization is in Table S8.

LOCALIZATION METHODOLOGY

Per mode, from $u(\boldsymbol{r}) = \varepsilon|\boldsymbol{E}|^2$ on the G^3 grid (`eigenfns/localization.py`): participation ratio $\mathrm{PR} = (\sum u)^2/\sum u^2$ (voxels), participation fraction $p = \mathrm{PR}/G^3$, participation volume $V_p = \mathrm{PR}\,\delta x^3$ and radius $r_p = (3V_p/4\pi)^{1/3}$. Envelope decay length: azimuthal average of u in 48 radial bins of minimum-image distance from the peak voxel up to $L/2$; least-squares line through $\ln u(r)$ over $[0.10, 0.95]\,L/2$; $u \propto e^{-2r/\xi}$, so slope $= -2/\xi$. A fit is flagged *unresolved* (lower bound only, never "extended, $\xi = X$") if $\xi \geq L/2$, if the profile decays by < 1 decade over the range, or if $r^2 < 0.7$ (non-exponential shape); thresholds pre-registered. $N = 10^4$: 121/133 resolved; the 12 unresolved all fail $r^2 \geq 0.7$, one also exceeds the ceiling. $N = 1000$ circular: 42/210 resolved (141 above the 5.72 μm ceiling, 77

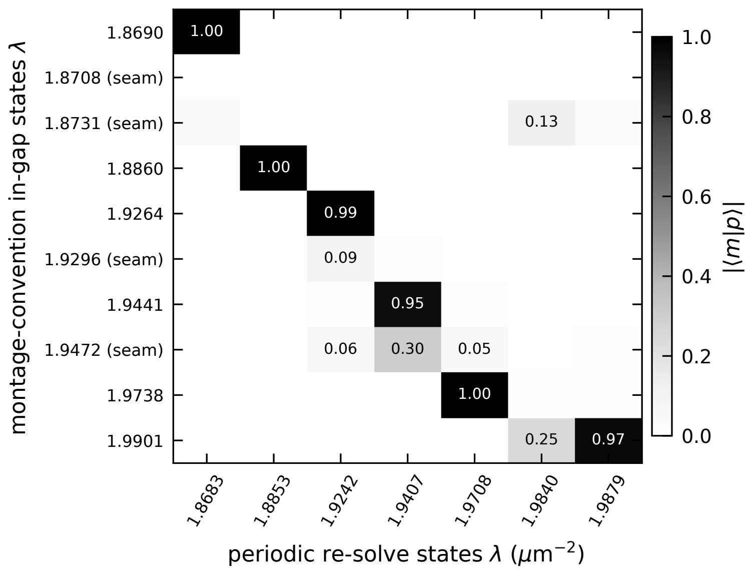

FIG. S8. Seam test: $|\langle m|p\rangle|$ between the ten montage-convention in-gap states (rows) and the seven periodic-solve states (columns), $m = 30$ run.

fail r^2, overlapping); elliptical: 30/210.

Fit-range sensitivity. Refitting over $[0.10, 0.60]\,L/2$ changes ξ by a median $|\Delta\xi|/\xi = 30\%$ over the 121 resolved $N = 10^4$ modes (compact modes, $p < 0.2\%$: median ratio 0.75, range 0.49–1.16; extended, $p > 1\%$: 0.66, range 0.48–1.64; Fig. S9); far radial bins of extended modes catch other hot spots. Medians by spectral bin (resolved modes): $[1.757, 1.80]$ 5.00, $[1.80, 1.85]$ 3.01, $[1.85, 1.864]$ 2.08, bracket 2.09, $[1.996, 2.02]$ 2.79, $[2.02, 2.07]$ 4.75, $[2.07, 2.117]$ 6.11 μm; with the short range 3.64, 2.32, 1.72, 1.76, 2.05, 3.66, 3.79 μm. Window-end/near-edge contrasts: ξ 1.7× (below) and 2.2× (above) with the full range, 1.8×/1.6× with the short range; r_p 1.8×/1.7× (r_p by bin: 2.65, 1.49, 1.15, 1.35, 3.12, 4.13, 5.43 μm); median p by bin 0.52, 0.09, 0.04, 0.07, 0.85, 2.0, 4.5%. Hot-spot spacing is flat at $\sim 1\,\mu$m (the rod scale) across all modes while ξ varies 7×, so ξ is not measuring hot-spot spacing. *I8* (cross-solver ξ agreement, $\leq 0.1\%$) shows the pipeline is deterministic, not that the estimator is accurate: both solvers hand it the same field to $\sim 10^{-7}$.

LEVEL STATISTICS

Adjacent-gap ratio $r_n = \min(s_n, s_{n-1})/\max(s_n, s_{n-1})$ [12]; surmises $P_{\mathrm{P}}(r) = 2/(1+r)^2$ and $P_{\mathrm{GOE}}(r) = \frac{27}{8}(r + r^2)/(1 + r + r^2)^{5/2}$, $\langle r\rangle_{\mathrm{P}} = 2\ln 2 - 1 = 0.3863$, $\langle r\rangle_{\mathrm{GOE}} = 0.5307$

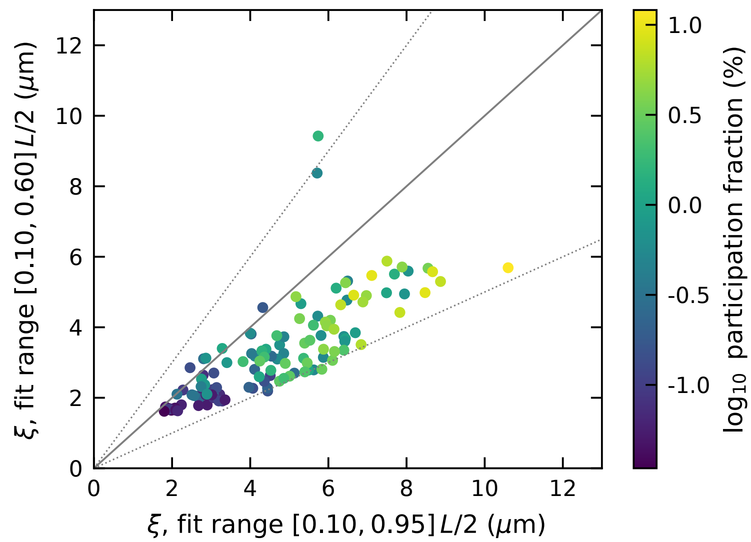

FIG. S9. Fit-range sensitivity of ξ for the 121 resolved $N = 10^4$ modes; dotted lines at 0.5× and 1.5×.

(large-N; the 3×3 surmise gives 0.5359) [13]. Bootstrap standard errors (20 000 resamples). Bands and results (Table S9, Fig. S10). Spacing distributions after local unfolding (9-level running mean) are shown for completeness; the ratio statistic is the one used. No pair of the 133 $N = 10^4$ levels is closer than 5% of the local spacing (Poisson expects 4.9%, GOE 0.4%); 1.8% / 1.0% of the 611 elliptical/circular $N = 1000$ spacings are, reflecting accidental near-degeneracies that gate G4 resolved as subspaces. Log-likelihood ratios GOE vs Poisson from the surmises are strongly in favour of GOE for every band with ≥ 40 levels (≥ 27 nats) and marginal for the 12-level near-edge band (-1.5... see `report/tables/levelstats.json`). The 7 states of the periodic re-solve are too few for any statistic. A Thouless-type sensitivity (twisted boundary conditions or dE/dk) would need re-solves at $k \neq \Gamma$ and is not available.

RETRACTED AND WITHDRAWN CLAIMS

The following statements were made during the project and are *not* made in this paper. (1) *Rare-region z-test* (retracted, round 4): the energy-weighted local filling fraction of the five candidates, z-scored against the ξ-coarse-grained field, gave $|z| = 0.57$ vs 0.27 for bulk controls. The controls were the most extended modes of the window ($p = 0.7$–12%) against the most compact candidates ($p = 0.03$–0.3%), a 36× median volume difference, and energy-weighting a coarse-grained field is a shrinkage estimator, so the ratio was manufactured;

TABLE S9. Adjacent-gap ratios (bootstrap s.e.). Poisson 0.386, GOE 0.531.

spectrum	band	λ range	levels	$\langle r \rangle$
$N = 10^4$	below, extended	< 1.816	51	0.528 ± 0.038
$N = 10^4$	below, near edge	1.821–1.850	12	0.357 ± 0.073
$N = 10^4$	in bracket (all ten)	1.869–1.990	10	0.314 ± 0.081
$N = 10^4$	above, near edge	2.005–2.050	18	0.559 ± 0.047
$N = 10^4$	above, extended	> 2.051	42	0.528 ± 0.041
$N = 10^4$	below bracket (all)	< 1.864	63	0.498 ± 0.035
$N = 10^4$	above bracket (all)	> 1.996	60	0.541 ± 0.032
$N = 10^4$	outside bracket, pooled r		119 r	0.519 ± 0.024
$N = 10^4$	all		133	0.501 ± 0.023
$N = 10^3$ ell.	bands 3–200		198	0.486 ± 0.019
$N = 10^3$ ell.	bands 201–400		200	0.495 ± 0.019
$N = 10^3$ ell.	bands 401–500		100	0.526 ± 0.026
$N = 10^3$ ell.	bands 501–600		100	0.542 ± 0.026
$N = 10^3$ ell.	all 611		611	0.508 ± 0.010
$N = 10^3$ circ.	bands 401–500		100	0.516 ± 0.028
$N = 10^3$ circ.	bands 501–600		100	0.560 ± 0.026
$N = 10^3$ circ.	all 611		611	0.533 ± 0.011

against an extent-matched translation null $|z|/\sigma_{\text{null}}$ is 2.77 for the candidates vs 7.19 for the controls ($p = 0.999$ in the claimed direction). Only the finite-size statistics argument (matched local distributions; 10× the draws; $P(\text{empty } N = 1000 \text{ gap}) = 0.61$) survives. (2) *"The resolution confound is closed"* (withdrawn, round 4): the 99.98% power retention of 256^3 modes inside the 192^3 k-set is saturated—a 96^3 cube retains 99.8% and the statistic is flat (99.975–99.984%) across all 14 modes—and it measures band-limiting of E, not of ε. The I6 gate passes on its per-band criterion; that is all that is claimed. (3) *"All four seam states vanished"*: 1.9472 is undetermined (Sec.), and the periodic solve is incomplete. (4) *Uncorrected eigenvalues* and every figure derived from them (I1 at 2.8×10^{-5}; $\text{proj}^2 > 1$). (5) *KPM/eigensolver in-gap agreement as evidence against an artefact* (11.18 ± 0.58 vs 10): KPM consumes the same rasterized ε and sees the same seam; it rules out an eigensolver

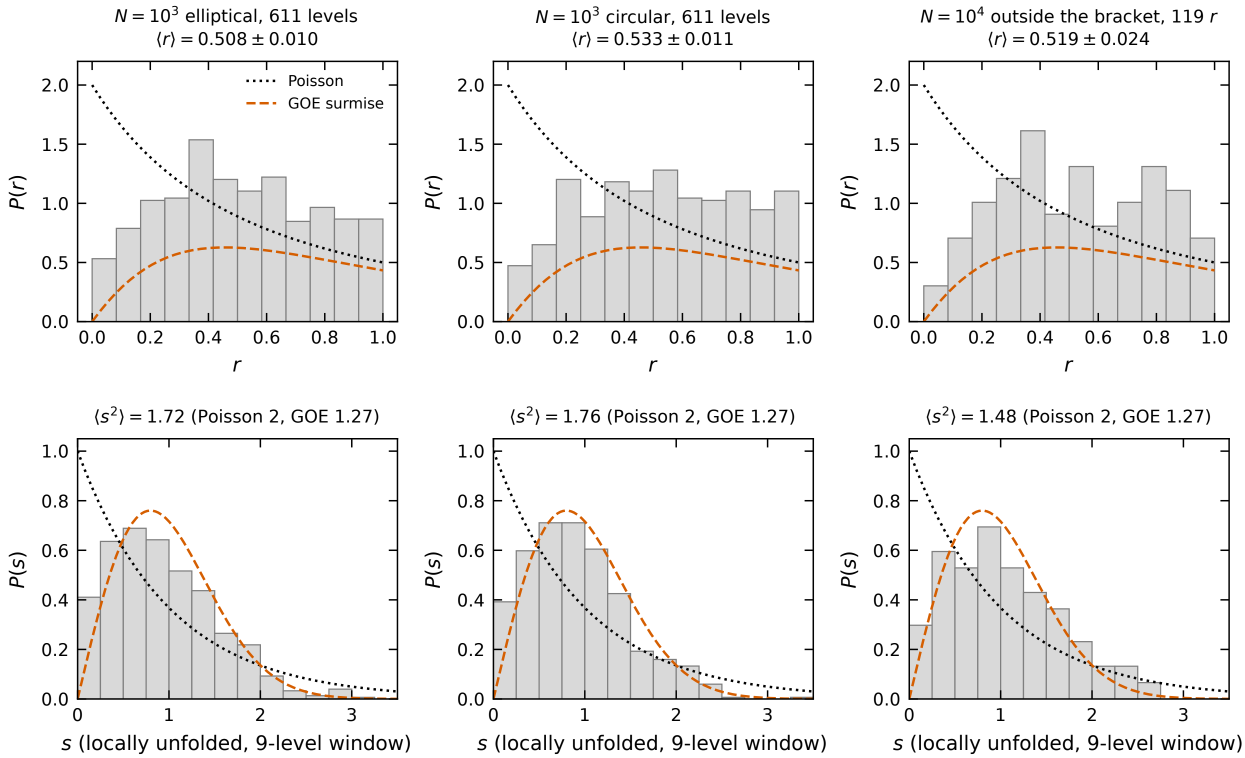

FIG. S10. $P(r)$ (top) and locally unfolded $P(s)$ (bottom) for the two $N = 1000$ spectra and the $N = 10^4$ levels outside the bracket, with the Poisson and GOE surmises.

artefact only, and carries a +1 bias. (6) *"S_{below} 69 predicted vs 69 found"* as a validation: with the +7.7 bias model the prediction is ≈ 65; the agreement was a coincidence. (7) The extended in-gap state 1.9441 was briefly listed as a localized candidate ($\xi = 13.0\,\mu\text{m} > L/2$, $r^2 = 0.33$). (8) Stale ranges in early drafts ("$\xi = 1.8$–$2.1\,\mu$m, $p = 0.03$–0.09%", "119 of 130 resolved") superseded by 1.80–2.51, 0.034–0.33%, 121 of 133.

HONEST LIMITATIONS (FULL LIST)

1. Gap-edge frequencies are rasterization-limited: 0.27% non-monotone spread at $N = 1000$ ($64^3/96^3/128^3$); the gap width is non-monotone (2.35/1.93/2.08%); on the MPB-interpolated 64^3 grid the elliptical 500|501 spacing is 0.60% with the largest spacing at 501|502. Sub-percent gap-edge frequencies are not claimed at 192^3 (7.8 vox/μm), where the 192/256 shifts are $\leq 0.34\%$.

2. Binary voxels (no subpixel smoothing) by design, for fidelity to the reference montage;

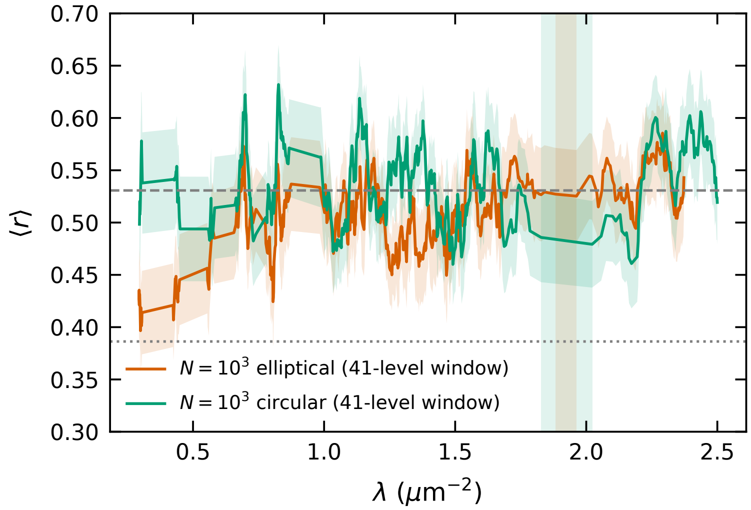

FIG. S11. Sliding 41-level $\langle r \rangle$ for the two $N = 1000$ spectra; shaded: the gaps.

a smoothed-ε convention would be a different calculation.

3. The montage-convention rasterizer carries the boundary seam (11–13% shell deficit) on all grids; the $N = 1000$ windows contain modes with shell enhancement up to 4×; only the $N = 10^4$ gap window was re-solved seam-free, and that re-solve is incomplete ($+2.0 \pm 0.4$ missed in its sub-interval; two solves left different states unconverged).

4. Completeness is certified on the sub-interval $[1.9063, 1.9606]$ only; the full-window estimator is biased and the KPM population comparison is not a count.

5. I3 fails: the merged 133-mode set has cross-slice Gram deviations up to 4.9×10^{-4}; mechanism open.

6. ξ is bounded by $L/2$, fit-range dependent at the 30% level, and validated only for pipeline determinism (I8), not as an absolute estimator; 168/210 $N = 1000$ modes are ceiling-limited.

7. One disorder realization per size; two sizes; no transport; no $k \neq \Gamma$ data (G1 not run).

8. fp32 device arithmetic with fp64 Rayleigh–Ritz; the normalization bug shows how such pipelines fail silently.

9. G4 was scope-limited to the lowest 30 bands; window degeneracies are certified only indirectly (parity 3.5×10^{-5}, Gram 1.2×10^{-5}).

10. Absolute $N = 10^4$ band indices carry ± 13 (stochastic) and a ≈ -4.5 Jackson bias; the ± 2 convention ambiguity against the original montage remains.

11. The 5.5 h $N = 1000$ production missed its 4 h target; the I6 high-edge anchor certified only 3 of 11 pairs; the 160^3 leg is edge-truncated; the first 256^3 high-edge attempt (9.1 h) produced nothing.

12. "No published 3-D supercell interior Maxwell eigensolver on GPU was found" is stated as "none found", not as a first.

TABLES

Table S12 lists all 133 certified $N = 10^4$ states. The 210-row $N = 1000$ window tables (both decorations) and all 611-band spectra are in the repository as `report/tables/states_n1k_{ell,cir` and `spectrum_n1k_{ell,circ}.csv`; Table S10 shows the circular-decoration bands around the gap.

TABLE S12: All 133 residual-certified $N = 10^4$ eigenstates (corrected $\lambda = \lambda_{\mathrm{raw}}/\|x\|^2$). $\nu = \sqrt{\lambda}\, a/2\pi$ with $a = 2.288\,\mu$m. Residuals are the reported (raw-λ) values. p is the participation fraction; ξ the envelope decay length ('u' = unresolved: lower bound only; ceiling $L/2 = 12.32\,\mu$m); f_2 the energy fraction in the outer 2-voxel shell (volume fraction 6.1%). Class: g = inside the nominal bracket [1.864, 1.996]; seam = boundary-seam artefact (seam* = undetermined, see text); ext. = extended in-gap mode. Periodic: best-overlap partner in the seam-free re-solve (in-gap states only).

#	tile	λ (μm^{-2})	ν	res.	p (%)	ξ (μm)	r^2	f_2 (%)	slice	class	periodic
0	4942	1.75705	0.4827	6.1×10^{-5}	1.294	6.20	0.90	5.8	Sbelow		
1	4943	1.75739	0.4827	6.2×10^{-5}	0.899	5.30	0.95	6.4	Sbelow		
2	4944	1.75835	0.4829	6.1×10^{-5}	0.717	7.95	0.73	8.8	Sbelow		
3	4945	1.75917	0.4830	6.1×10^{-5}	1.286	7.87u	0.62	5.8	Sbelow		
4	4946	1.76006	0.4831	6.1×10^{-5}	1.358	7.69	0.84	7.1	Sbelow		
5	4947	1.76054	0.4832	6.0×10^{-5}	1.073	7.49	0.83	3.0	Sbelow		
6	4948	1.76138	0.4833	6.1×10^{-5}	0.608	6.40	0.78	4.8	Sbelow		
7	4949	1.76157	0.4833	6.0×10^{-5}	0.667	4.75	0.89	7.5	Sbelow		
8	4950	1.76246	0.4834	6.0×10^{-5}	0.567	6.48	0.85	3.6	Sbelow		
9	4951	1.76369	0.4836	6.0×10^{-5}	0.238	4.02	0.94	6.0	Sbelow		
10	4952	1.76504	0.4838	6.2×10^{-5}	0.694	8.04	0.77	5.5	Sbelow		
11	4953	1.76516	0.4838	6.2×10^{-5}	1.014	6.44	0.78	9.2	Sbelow		

TABLE S10. $N = 1000$, circular decoration, bands 494–507 at 128^3 (^{u} = unresolved fit, lower bound).

band	λ (μm^{-2})	ν	p (%)	V_p (μm^3)	ξ (μm)	r^2
494	1.75286	0.4821	2.74	41.1	2.58	0.87
495	1.76650	0.4840	2.86	42.9	3.96^{u}	0.61
496	1.76737	0.4841	3.91	58.5	4.82^{u}	0.65
497	1.77580	0.4853	2.96	44.4	3.52^{u}	0.68
498	1.77759	0.4855	1.33	19.9	2.94	0.84
499	1.78550	0.4866	0.59	8.8	1.73	0.94
500	1.82759	0.4923	0.42	6.2	1.47	0.97
501	2.02249	0.5179	3.86	57.8	2.30	0.95
502	2.05698	0.5223	10.82	162.0	4.14	0.96
503	2.07574	0.5246	13.24	198.2	3.41	0.96
504	2.10055	0.5278	11.25	168.4	3.50	0.88
505	2.11251	0.5293	19.80	296.4	12.70^{u}	0.89
506	2.12920	0.5314	21.00	314.4	4.72^{u}	0.93
507	2.13615	0.5322	22.23	332.8	4.73^{u}	0.96

#	tile	λ	ν	res.	p (%)	ξ	r^2	f_2 (%)	slice	class	periodic
12	4954	1.76540	0.4838	6.0×10^{-5}	0.754	6.42	0.76	8.2	Sbelow		
13	4955	1.76571	0.4839	6.1×10^{-5}	0.619	5.81	0.81	11.0	Sbelow		
14	4956	1.76656	0.4840	6.1×10^{-5}	0.879	6.69	0.75	6.4	Sbelow		
15	4957	1.76754	0.4841	6.1×10^{-5}	0.527	4.12	0.95	6.0	Sbelow		
16	4958	1.76810	0.4842	5.9×10^{-5}	0.536	5.87	0.73	7.7	Sbelow		
17	4959	1.76892	0.4843	6.2×10^{-5}	0.389	4.03	0.92	10.2	Sbelow		
18	4960	1.76910	0.4843	6.2×10^{-5}	0.482	4.86	0.84	10.0	Sbelow		
19	4961	1.77082	0.4846	5.9×10^{-5}	0.234	4.11	0.89	2.5	Sbelow		
20	4962	1.77127	0.4846	5.9×10^{-5}	0.481	8.66u	0.51	4.9	Sbelow		
21	4963	1.77420	0.4850	5.9×10^{-5}	0.522	5.71	0.91	13.4	Sbelow		
22	4964	1.77656	0.4854	5.9×10^{-5}	0.594	5.73	0.86	4.2	Sbelow		
23	4965	1.77677	0.4854	6.0×10^{-5}	0.453	6.49	0.82	5.5	Sbelow		
24	4966	1.77858	0.4856	5.9×10^{-5}	0.143	4.07	0.79	1.9	Sbelow		
25	4967	1.77929	0.4857	5.9×10^{-5}	0.153	4.45	0.71	0.8	Sbelow		
26	4968	1.78053	0.4859	6.1×10^{-5}	0.562	4.69	0.88	1.2	Sbelow		
27	4969	1.78157	0.4860	6.0×10^{-5}	0.577	4.00	0.98	11.5	Sbelow		
28	4970	1.78291	0.4862	6.1×10^{-5}	0.534	5.63	0.71	8.4	Sbelow		
29	4971	1.78511	0.4865	6.2×10^{-5}	0.105	4.51	0.78	1.4	Sbelow		
30	4972	1.78599	0.4866	6.4×10^{-5}	0.064	2.69	0.95	0.7	Sbelow		
31	4973	1.78791	0.4869	6.1×10^{-5}	0.172	4.32	0.88	11.2	Sbelow		
32	4974	1.78938	0.4871	6.2×10^{-5}	0.466	4.86	0.92	6.4	Sbelow		
33	4975	1.79004	0.4872	6.3×10^{-5}	0.279	5.13	0.79	3.1	Sbelow		
34	4976	1.79097	0.4873	6.2×10^{-5}	0.136	3.07	0.97	3.4	Sbelow		
35	4977	1.79460	0.4878	6.1×10^{-5}	0.071	3.22	0.90	6.2	Sbelow		
36	4978	1.79596	0.4880	6.0×10^{-5}	0.293	2.81	0.98	16.8	Sbelow		

TABLE S11. $N = 1000$, elliptical decoration, bands 494–507 at 128^3.

band	λ (μm^{-2})	ν	p (%)	V_p (μm^3)	ξ (μm)	r^2
494	1.79589	0.4880	2.31	34.6	2.58	0.93
495	1.80177	0.4888	3.35	50.1	3.07	0.75
496	1.80684	0.4895	1.54	23.0	2.18	0.90
497	1.81992	0.4913	6.06	90.7	6.43^{u}	0.50
498	1.82787	0.4923	4.48	67.1	4.92	0.73
499	1.84065	0.4940	0.86	12.8	2.05	0.92
500	1.88296	0.4997	0.81	12.2	1.82	0.97
501	1.96277	0.5102	7.82	117.1	2.92	0.99
502	1.98657	0.5132	7.19	107.7	2.36	0.99
503	2.01179	0.5165	12.11	181.3	3.50	0.99
504	2.02106	0.5177	17.28	258.8	4.29^{u}	0.97
505	2.04004	0.5201	16.44	246.1	5.53^{u}	0.79
506	2.04605	0.5209	18.03	269.9	6.19^{u}	0.96
507	2.05624	0.5222	9.63	144.2	3.08	0.98

#	tile	λ	ν	res.	p (%)	ξ	r^2	f_2 (%)	slice	class	periodic
37	4979	1.79713	0.4882	6.1×10^{-5}	0.465	8.57u	0.56	13.0	Sbelow		
38	4980	1.79765	0.4882	6.0×10^{-5}	0.359	4.56u	0.68	19.2	Sbelow		
39	4981	1.79793	0.4883	6.1×10^{-5}	0.347	6.14u	0.62	17.9	Sbelow		
40	4982	1.79885	0.4884	6.0×10^{-5}	0.220	3.97	0.80	3.9	Sbelow		
41	4983	1.80001	0.4886	6.2×10^{-5}	0.122	3.01	0.91	5.3	Sbelow		
42	4984	1.80045	0.4886	6.0×10^{-5}	0.261	4.54	0.86	3.7	Sbelow		
43	4985	1.80290	0.4889	6.0×10^{-5}	0.221	4.44	0.75	8.2	Sbelow		
44	4986	1.80362	0.4890	6.1×10^{-5}	0.230	2.70	0.93	16.3	Sbelow		
45	4987	1.80435	0.4891	6.4×10^{-5}	0.149	4.36	0.77	4.3	Sbelow		
46	4988	1.80530	0.4893	6.2×10^{-5}	0.097	3.04	0.91	3.5	Sbelow		
47	4989	1.80639	0.4894	6.2×10^{-5}	0.053	2.90	0.85	0.4	Sbelow		
48	4990	1.80754	0.4896	6.4×10^{-5}	0.207	3.06	0.94	27.9	Sbelow		
49	4991	1.81234	0.4902	6.0×10^{-5}	0.059	2.24	0.98	0.5	Sbelow		
50	4992	1.81592	0.4907	6.2×10^{-5}	0.093	2.28	0.98	1.6	Sbelow		
51	4993	1.82084	0.4914	6.0×10^{-5}	0.081	3.06	0.91	24.8	Sbelow		
52	4994	1.82172	0.4915	6.1×10^{-5}	0.080	3.28	0.86	15.1	Sbelow		
53	4995	1.82277	0.4916	6.1×10^{-5}	0.109	3.20	0.88	2.2	Sbelow		
54	4996	1.82312	0.4917	6.4×10^{-5}	0.055	2.68	0.91	4.4	Sbelow		
55	4997	1.82437	0.4919	6.0×10^{-5}	0.053	3.36	0.85	2.7	Sbelow		
56	4998	1.82960	0.4926	6.1×10^{-5}	0.057	2.87	0.93	6.0	Sbelow		
57	4999	1.82983	0.4926	5.9×10^{-5}	0.139	3.14	0.95	1.5	Sbelow		
58	5000	1.83504	0.4933	6.0×10^{-5}	0.078	2.80	0.95	18.5	Sbelow		
59	5001	1.83796	0.4937	6.0×10^{-5}	0.137	2.46	0.97	18.3	Sbelow		
60	5002	1.84478	0.4946	5.8×10^{-5}	0.066	2.03	0.98	0.7	Sbelow		
61	5003	1.84662	0.4948	6.1×10^{-5}	0.039	1.98	0.97	5.0	Sbelow		
62	5004	1.85015	0.4953	6.1×10^{-5}	0.042	2.08	0.97	5.6	Sbelow		
63	5005	1.86898	0.4978	5.9×10^{-5}	0.072	1.91	0.99	2.7	Sbelow	g (cand.)	1.8683 (0.999)

#	tile	λ	ν	res.	p (%)	ξ	r^2	f_2 (%)	slice	class	periodic
64	5006	1.87076	0.4981	5.9×10^{-5}	0.048	1.83	0.99	18.2	Sbelow	seam	– (0.006)
65	5007	1.87308	0.4984	6.1×10^{-5}	0.093	1.91	0.99	23.6	Sbelow	seam	– (0.131)
66	5008	1.88601	0.5001	6.0×10^{-5}	0.034	1.80	0.99	2.8	Sbelow	g (cand.)	1.8853 (1.000)
67	5009	1.92641	0.5054	5.7×10^{-5}	0.046	2.09	0.97	2.5	Sbelow	g (cand.)	1.9242 (0.994)
68	5010	1.92960	0.5058	6.3×10^{-5}	0.065	2.14	0.96	43.7	Sbelow	seam	– (0.090)
69	5011	1.94405	0.5077	5.5×10^{-5}	0.558	12.97u	0.32	8.1	Sgap	ext.	1.9407 (0.952)
70	5012	1.94721	0.5081	5.6×10^{-5}	0.053	3.01	0.92	42.1	Sgap	seam*	– (0.301)
71	5013	1.97381	0.5116	5.4×10^{-5}	0.329	2.13	1.00	10.5	Sgap	g (cand.)	1.9708 (0.997)
72	5014	1.99012	0.5137	5.2×10^{-5}	0.290	2.51	0.98	3.0	Sabove	g (cand.)	1.9879 (0.968)
73	5015	2.00521	0.5157	5.3×10^{-5}	0.339	2.56	0.96	0.7	Sabove		
74	5016	2.00769	0.5160	5.1×10^{-5}	0.919	2.75	0.99	2.7	Sabove		
75	5017	2.00879	0.5161	5.2×10^{-5}	0.789	2.84	0.95	12.4	Sabove		
76	5018	2.01185	0.5165	5.3×10^{-5}	0.418	2.74	0.97	19.6	Sabove		
77	5019	2.01468	0.5169	5.2×10^{-5}	1.145	5.49	0.77	1.6	Sabove		
78	5020	2.01637	0.5171	5.1×10^{-5}	1.167	4.40	0.92	1.6	Sabove		
79	5021	2.02093	0.5177	5.1×10^{-5}	0.789	3.40	0.99	14.7	Sabove		
80	5022	2.02738	0.5185	5.3×10^{-5}	1.971	5.62	0.92	8.7	Sabove		
81	5023	2.03088	0.5189	5.2×10^{-5}	1.605	5.74	0.94	16.0	Sabove		
82	5024	2.03292	0.5192	5.1×10^{-5}	1.962	3.82	0.97	6.2	Sabove		
83	5025	2.03675	0.5197	5.2×10^{-5}	2.057	4.68	0.97	5.0	Sabove		
84	5026	2.03965	0.5201	5.2×10^{-5}	0.660	2.89	0.93	20.2	Sabove		
85	5027	2.04057	0.5202	5.1×10^{-5}	0.717	2.87	0.99	0.6	Sabove		
86	5028	2.04245	0.5204	5.1×10^{-5}	4.795	6.97	0.88	8.8	Sabove		
87	5029	2.04526	0.5208	5.1×10^{-5}	1.161	3.28	0.98	1.7	Sabove		
88	5030	2.04616	0.5209	5.3×10^{-5}	1.785	5.37	0.82	5.1	Sabove		
89	5031	2.04712	0.5210	5.5×10^{-5}	3.580	8.55	0.83	3.7	Sabove		
90	5032	2.04952	0.5213	5.3×10^{-5}	1.490	4.30	0.95	5.2	Sabove		
91	5033	2.05075	0.5215	5.1×10^{-5}	2.789	4.75	0.78	4.6	Sabove		
92	5034	2.05228	0.5217	5.2×10^{-5}	3.628	6.05	0.85	3.1	Sabove		
93	5035	2.05373	0.5219	5.1×10^{-5}	4.267	7.89	0.92	5.1	Sabove		
94	5036	2.05458	0.5220	5.1×10^{-5}	1.984	4.36	0.93	3.0	Sabove		
95	5037	2.05686	0.5223	5.1×10^{-5}	1.148	4.23	0.87	9.3	Sabove		
96	5038	2.05765	0.5224	5.1×10^{-5}	4.102	6.44	0.94	4.8	Sabove		
97	5039	2.06035	0.5227	5.2×10^{-5}	2.365	6.40	0.79	1.9	Sabove		
98	5040	2.06442	0.5232	5.2×10^{-5}	4.786	11.02u	0.44	5.5	Sabove		
99	5041	2.06639	0.5235	5.3×10^{-5}	2.997	5.39	0.91	4.5	Sabove		
100	5042	2.06908	0.5238	5.4×10^{-5}	1.548	4.53	0.88	5.1	Sabove		
101	5043	2.07022	0.5239	5.3×10^{-5}	3.201	5.46	0.81	4.6	Sabove		
102	5044	2.07044	0.5240	5.3×10^{-5}	2.958	4.90	0.88	2.8	Sabove		
103	5045	2.07270	0.5243	5.1×10^{-5}	2.800	4.24	0.94	5.7	Sabove		
104	5046	2.07504	0.5246	5.2×10^{-5}	6.362	7.50	0.93	6.4	Sabove		
105	5047	2.07622	0.5247	5.1×10^{-5}	3.655	7.65u	0.68	4.2	Sabove		
106	5048	2.07704	0.5248	5.2×10^{-5}	4.660	5.92	0.91	3.1	Sabove		
107	5049	2.07867	0.5250	5.1×10^{-5}	6.107	6.84	0.78	5.4	Sabove		
108	5050	2.08010	0.5252	5.2×10^{-5}	3.429	6.11	0.77	4.1	Sabove		
109	5051	2.08059	0.5253	5.2×10^{-5}	4.452	5.96	0.92	5.2	Sabove		
110	5052	2.08201	0.5254	5.1×10^{-5}	3.183	5.83	0.73	3.5	Sabove		
111	5053	2.08464	0.5258	5.2×10^{-5}	2.789	5.01	0.79	2.9	Sabove		
112	5054	2.08537	0.5259	5.2×10^{-5}	3.485	5.46u	0.70	4.8	Sabove		
113	5055	2.08721	0.5261	5.2×10^{-5}	3.900	7.63u	0.53	4.3	Sabove		
114	5056	2.08938	0.5264	5.2×10^{-5}	2.008	4.86	0.79	3.6	Sabove		
115	5057	2.08980	0.5264	5.3×10^{-5}	2.390	5.40	0.75	3.2	Sabove		
116	5058	2.09186	0.5267	5.2×10^{-5}	2.622	5.53	0.89	5.8	Sabove		
117	5059	2.09342	0.5269	5.1×10^{-5}	7.178	7.83	0.80	4.5	Sabove		
118	5060	2.09473	0.5270	5.1×10^{-5}	7.073	8.87	0.80	5.4	Sabove		
119	5061	2.09679	0.5273	5.1×10^{-5}	8.666	8.67	0.88	6.1	Sabove		
120	5062	2.09806	0.5275	5.2×10^{-5}	5.411	6.14	0.86	6.9	Sabove		
121	5063	2.09911	0.5276	5.1×10^{-5}	9.304	10.91u	0.68	7.4	Sabove		
122	5064	2.09937	0.5276	5.2×10^{-5}	4.371	5.88	0.83	7.0	Sabove		
123	5065	2.10012	0.5277	5.2×10^{-5}	5.239	6.88	0.88	6.3	Sabove		
124	5066	2.10272	0.5280	5.2×10^{-5}	3.710	5.26	0.98	5.7	Sabove		
125	5067	2.10484	0.5283	5.2×10^{-5}	4.503	5.17	0.99	2.7	Sabove		
126	5068	2.10723	0.5286	5.1×10^{-5}	8.960	6.65	0.95	7.1	Sabove		

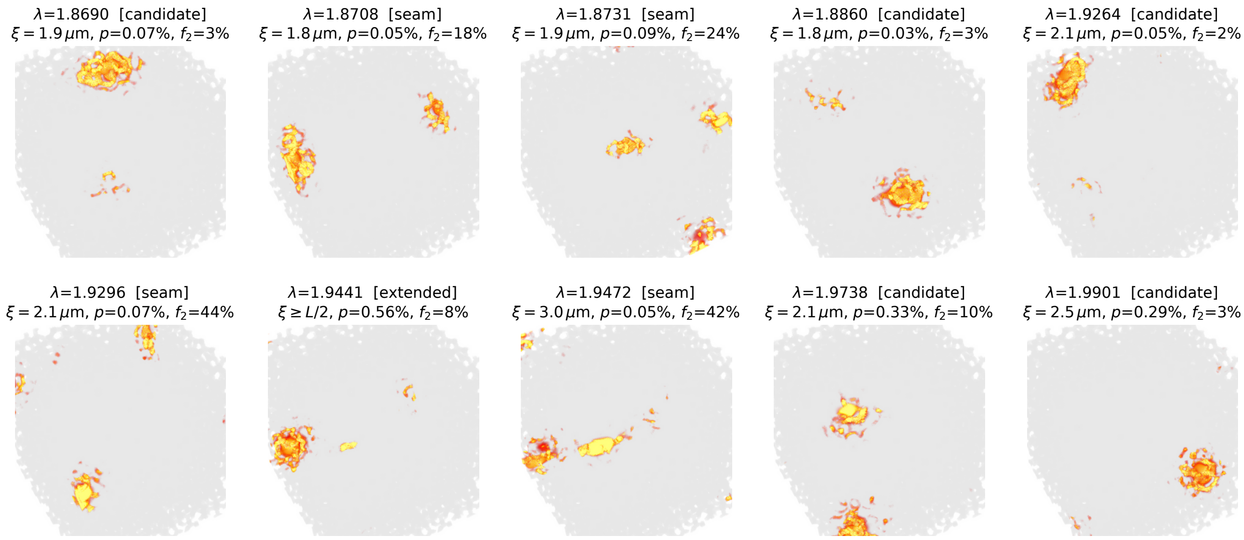

FIG. S12. The ten certified $N = 10^4$ states inside the nominal gap (montage-convention run), with class, ξ, p and outer-shell energy fraction f_2.

#	tile	λ	ν	res.	p (%)	ξ	r^2	f_2 (%)	slice	class	periodic
127	5069	2.10743	0.5286	5.1×10^{-5}	7.584	8.75u	0.66	7.0	Sabove		
128	5070	2.10954	0.5289	5.2×10^{-5}	12.082	10.60	0.73	5.1	Sabove		
129	5071	2.11240	0.5293	5.6×10^{-5}	6.958	6.32	0.90	3.1	Sabove		
130	5072	2.11383	0.5294	8.7×10^{-5}	9.603	7.11	0.96	5.6	Sabove		
131	5073	2.11503	0.5296	6.1×10^{-5}	3.770	6.16	0.81	3.4	Sabove		
132	5074	2.11657	0.5298	6.2×10^{-5}	9.293	8.47	0.81	5.7	Sabove		

EXTENDED FIGURES

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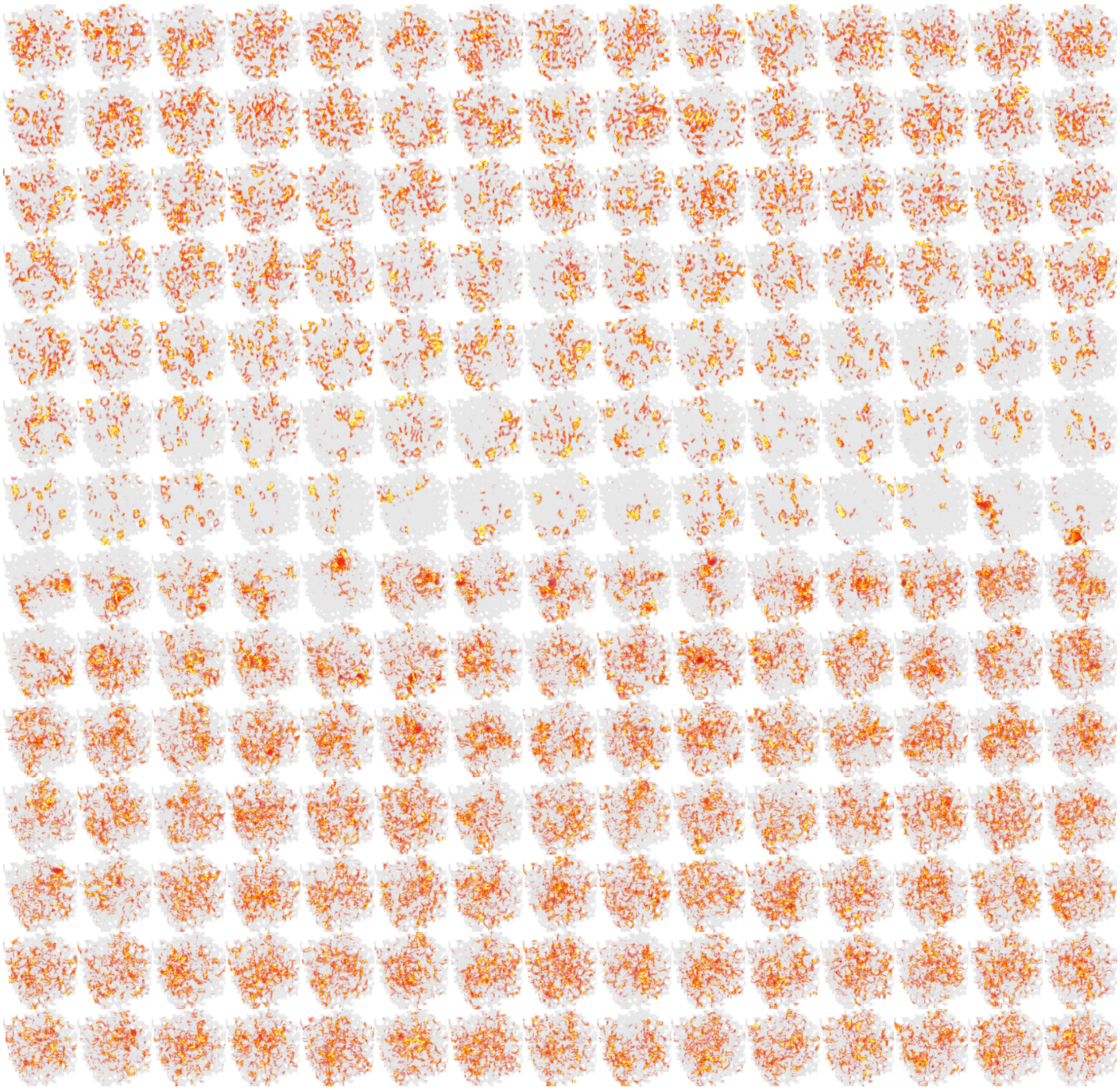

FIG. S13. $N = 1000$, elliptical decoration: the regenerated 210-tile montage of bands 398–607 (15 per row, row-major), $\varepsilon|\boldsymbol{E}|^2$ at 128^3.

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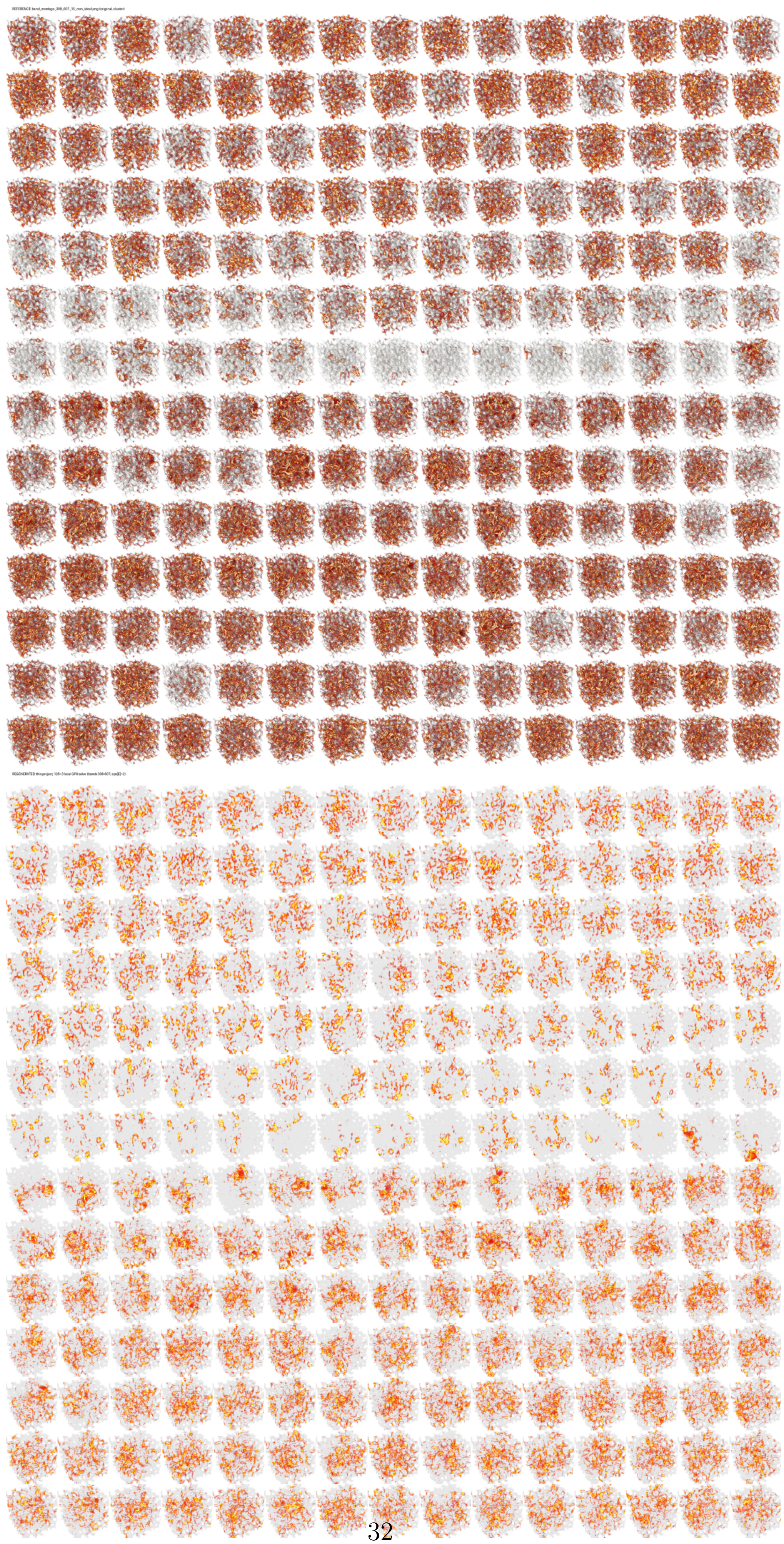

FIG. S14. Left: the original cluster montage `band_montage_398_607_15_non_ideal.png`; right:

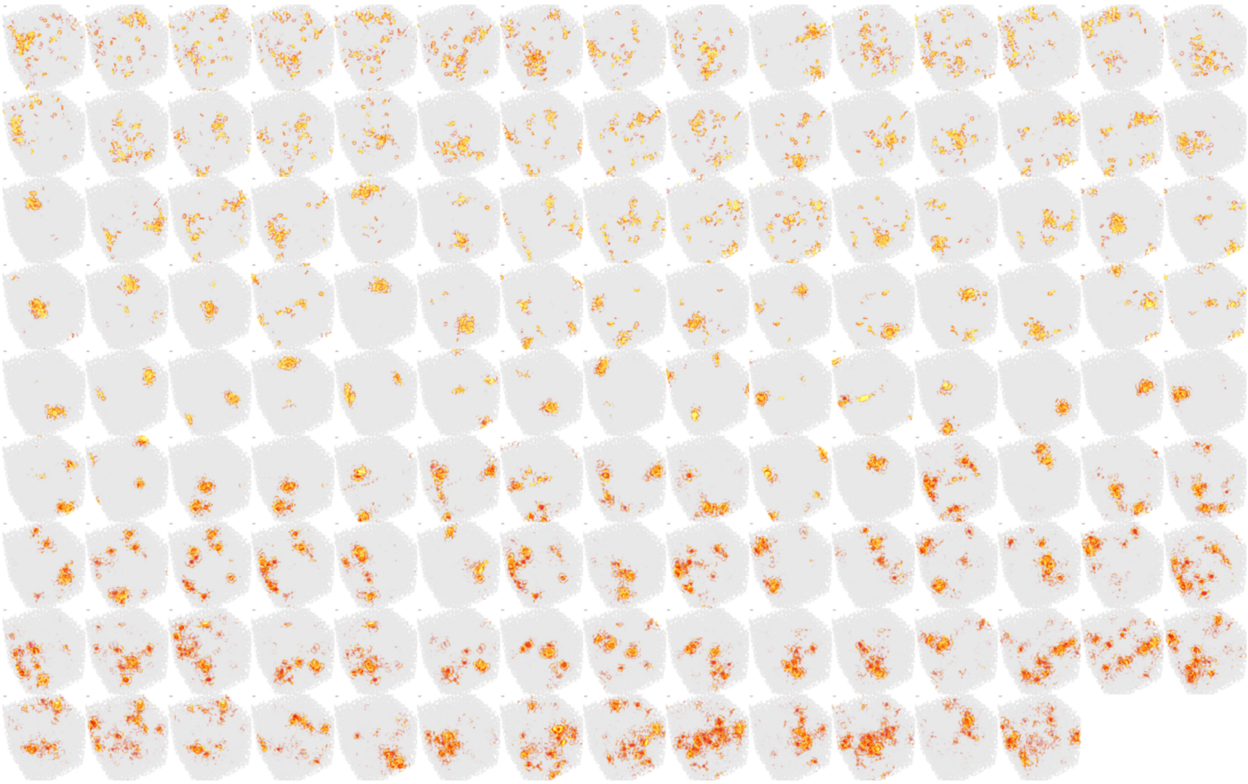

FIG. S15. $N = 10^4$: the 133 certified gap-edge modes, 15 per row, in increasing λ (labels 4942–5074, see text for the index caveat). Read top to bottom: energy spread over the box (window bottom), collapse to single compact blobs at the gap edges and inside the gap, spreading out again above.

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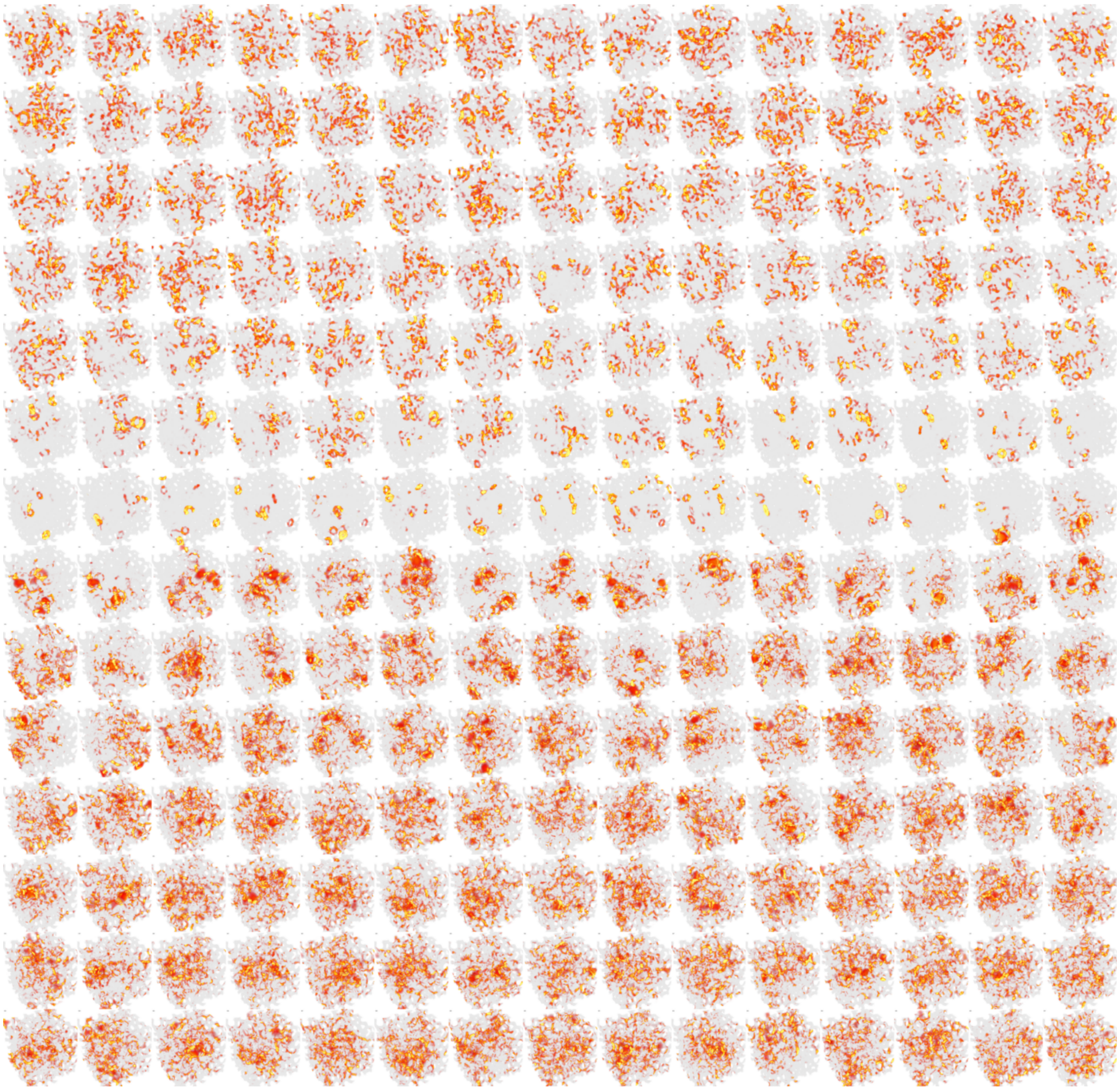

FIG. S16. $N = 1000$, circular decoration: 210-tile montage of bands 398–607 for the finite-size comparison with Fig. S15 (same decoration).